Grothendieck's Dessins d'Enfants

Translation of Master's Thesis

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Contents

1 Introduction **3**

2 Preliminaries **3**

2.1 Compact Riemann Surfaces and Algebraic Curves 4

2.2 Galois Theory 13

2.3 Galois Correspondance in Algebraic Topology 20

3 Belyi's Theorem **21**

4 Definition of Dessins **23**

4.1 Topological Defenition 23

4.2 Group Theory Definition 26

5 Grothendieck's Correspondence **28**

6 Galois Action on Dessins **30**

6.1 The Action and Faithfullness 30

6.2 Galois Invariants 31

6.2.1 Elementary Invariants 31

6.2.2 Valency Type 32

6.2.3 Monodromy Groups and Composition 33

6.2.4 Trees 33

7 Explicit Examples **35**
7.1 A Genus 0 Example 35
7.2 A Genus 1 Example 36
7.3 Two Examples From LMFDB 38

8 Two Applications in Number Theory **39**
8.1 ABC Conjecture 39
8.2 Elliptic Curves and Complex Multiplication 41

9 Some Research Directions **43**
9.1 Computation on Belyi maps and Dessins 43
9.2 Grothendieck-Teichmuller Theory 44

1 Introduction

In his 1984 proposal "Esquisse d'un programme" (Sketch of a program), Alexander Grothendieck, the famous French mathematician of the 20th century, studied a group of embedded bipartite graphs on surfaces which were called "dessin d'enfants" (it's word by word translation is "children's drawings"). This program was originally influenced by Belyi's theorem (from 1979 [Bel80]) on finding a necessary and sufficient condition for a compact Riemann Surface (algebraic curves over complex numbers) to be defined over the field of algebraic numbers $\bar{\mathbb{Q}}$.
One of the major importances of these graphs is a faithful action from the Absolute Galois Group of rationals on dessins, which opens combinatorial ways to study this mysterious group. Finding the Galois invariants of this action is as hard (and important!) as studying the absolute Galois group. Sections 3 to 6 of this note, explain these ideas.
This note is the translation of my Master's thesis at the University of Tehran. My thesis also had an epilogue, which was about a possible future research on "Grothendieck-Teichmuller" theory. This theory which was started by Ihara and Drinfeld, is a continuation of Grothendieck's ideas in "Sketch of a program" in Teichmuller spaces. I will explain these ideas in the last section of this note breifly. This great theorem of Ihara is somehow based on the modern view to the Galois theory which was again, one of the exciting works of Grothendieck in his famous SGA series [GR71]. This topic will be discussed briefly in the last part of "Preliminaries" and hopefully, I will add two more notes on my GitHub page to explain the details of these topics which I read after my Master's.

2 Preliminaries

In this section, we will review some important facts on compact Riemann surfaces, algebraic curves and Galois theory. The first part is essentially from [JW16]. The Galois theory part is a bit longer to discuss infinite dimensional Galois theory and is mainly from [Mil22] and [Lan05]. In the last part, I will explain (a little!) about Grothendieck's approach to Galois theory, using abstract notions of category theory. Two exciting books for this part are [Sza09] and [BJ01]. I will not discuss the details of category theory in this part to keep this note short but it is worth mentioning that one of the ultimate goals of these notes is to explain Ihara's work on the embedding $Gal(\bar{\mathbb{Q}}/\mathbb{Q}) \hookrightarrow \widehat{GT}$ which is highly based on the categorical view to Galois theory (especially in the context of Schemes in algebraic geometry and their algebraic fundamental group).
As this part is mainly for reviewing, almost all of the proofs are skipped. The reader can find the datails of the proofs in almost any standad book on these subjects or the mentioned references.

2.1 Compact Riemann Surfaces and Algebraic Curves

We start with the (not so formal!) defintion of a Riemann surface. In this note we consider them connected:

Definition 1. *Riemann surfaces are Hausdorff togological spaces which have a countable basis for their topology. They are (smooth) manifolds whose chart maps (local coordinates) take their values in the complex plane and are defined with biholomorphic transition functions where their domains overlap.*

Let's see some basic examples of this definition:

Example 1. *First example is the Riemann sphere, or sometimes called the complex projective line:* $\hat{\mathbb{C}} = \mathbb{P}^1(\mathbb{C}) = \mathbb{C} \cup \{\infty\}$.
We take two open subsets, for example $U_1 = \mathbb{C}$ *and* $U_2 = (\mathbb{C}\backslash\{0\}) \cup \{\infty\}$*, with chart maps* $U_1 \to \mathbb{C}, z \mapsto z$ *and* $U_2 \to \mathbb{C}, z \mapsto 1/z$ *(where, by convention, we interpret* $1/\infty$ *as 0). Then* $z \mapsto 1/z$ *is a biholomorphic transition function between local coordinates on the intersection* $U_1 \cap U_2 = \mathbb{C}\backslash\{0\}$*. We therefore have a Riemann surface. It can be identified, by stereographic projection, with the unit sphere in euclidean 3-space, so it is compact.*

Example 2. *The Fermat curve of degree* $n > 1$ *(as an affine curve) is defined to be:*

$$F_n^{aff} := \left\{(x, y) \in \mathbb{C}^2 \mid x^n + y^n = 1\right\}.$$

We can take as chart maps $(x, y) \mapsto y$*, which is a homeomorphism on suitable neighbourhoods of all points except those where* $x = 0$ *and* $y^n = 1$*, and* $(x, y) \mapsto x$ *which behaves similarly except where* $y = 0$ *and* $x^n = 1$*. As transition functions we take holomorphic branches of* $x = \sqrt[n]{1 - y^n}$ *and* $y = \sqrt[n]{1 - x^n}$*.*
Unlike $\hat{\mathbb{C}}$*, this Riemann surface is not compact: for instance, the continuous real-valued function* $(x, y) \mapsto |x|$ *is unbounded on* F_n^{aff} *.*

Example 3. *More generally we can take any smooth affine algebraic curve:*

$$X^{aff} := \left\{(x, y) \in \mathbb{C}^2 \mid f(x, y) = 0\right\}$$

where f *is a polynomial such that at each point* $p \in X^{aff}$ *either*

$$\frac{\partial f}{\partial x}(p) \neq 0 \quad \text{or} \quad \frac{\partial f}{\partial y}(p) \neq 0$$

The implicit function theorem implies that locally around p *all solutions of* $f(x, y) = 0$ *are of the form* $(h(y), y)$ *or* $(x, g(x))$ *respectively where* h *and* g *are holomorphic. Then the projections onto the coordinates* y *or* x *can be used as chart maps.*

Example 4. *As a special case of the previous example, an affine hyperelliptic curve is given by an equation*

$$y^2 = (x - a_1) \dots (x - a_n)$$

with pairwise distinct $a_1, \dots, a_n \in \mathbb{C}$. *Taking* $f(x, y) = y^2 - \prod_j (x - a_j)$ *we have*

$$\frac{\partial f}{\partial y}(p) = 2y = 0$$

only at the points $p = p_j = (a_j, 0)$, *so away from these points we can use* x *as a local coordinate. At the points* $p = p_j$ *we have*

$$\frac{\partial f}{\partial x}(p) \neq 0,$$

so near them we can use y *as a local coordinate instead.*

The chart maps allow us to define holomorphic and meromorphic functions on Riemann surfaces and holomorphic mappings between Riemann surfaces by tracing back all these properties locally to the usual definitions in domains of the complex plane. Thus holomorphic and meromorphic functions on Riemann surfaces inherit the usual properties from holomorphic and meromorphic functions in the plane.

Proposition 1. *Let* $f : X \to Y$ *be a holomorphic function between compact (and connected) Riemann surfaces. The open mapping theorem (for non-constant holomorphic functions) implies* $f(X) \subset Y$ *is a open set. Since* f *is continuous and* X *is compact,* $f(X) \subset Y$ *should also be compact (and so closed) in* Y. *So* $f(X) = Y$ *and* f *should be surjective. If* $Y = \mathbb{C}$ *then this contradiction shows* f *must be constant.*

Proposition 2. *Let* $f : X \to Y$ *be a non-constant holomorphic mapping between connected Riemann surfaces* X *and* Y, *let* $p \in X$, *and let* $p' = f(p)$. *Then there exist chart maps* $z : U(p) \to V \subset \mathbb{C}$ *and* $w : U'(p') \to V' \subset \mathbb{C}$ *with* $z(p) = 0$ *and* $w(p') = 0$, *and an integer* $n \in \mathbb{N}$ *such that the diagram*

$$\begin{tikzcd} U \arrow[r] \arrow[d, "z"'] & U' \arrow[d, "w"] \\ z \in \mathbb{C} \arrow[r, mapsto] & w = z^n \in \mathbb{C} \end{tikzcd}$$

is commutative. This integer n, *which is independent of the choice of the charts, is called* ***the 'multiplicity' mult*** $_p f$ ***of*** f ***at*** p.

If $n = 1$ then f is locally biholomorphic (unramified at p); otherwise it is ramified with order $n > 1$. Here are some consequences:

1. If $f : X \to \hat{\mathbb{C}}$ is meromorphic and non-constant, then its zeros and poles form a discrete subset of X.

2. The ramification points of f form a discrete subset of X.

3. The identity theorem, the maximum principle, and the open mapping theorem, familiar from complex function theory for domains in the complex plane, are also valid on Riemann surfaces.

4. If X is a compact Riemann surface then a non-constant meromorphic function $f : X \to \hat{\mathbb{C}}$ has only a finite number of zeros or poles, and also only a finite number of ramification points. A holomorphic function $f : X \to \mathbb{C}$ must be constant.

5. If X is a compact Riemann surface then any non-constant holomorphic function $f : X \to Y$ is surjective, and Y is also compact.

6. Under the same hypotheses the degree

$$\deg f := \sum_{p \in f^{-1}(y)} \operatorname{mult}_p f$$

of f is independent of the choice of $y \in Y$.

Before continuing with more basic facts of Riemann surfaces, we give some more examples.

Example 5. *Fermat curves again: the projective version of the Fermat curve of degree n is*

$$F_n := \left\{ [x, y, z] \in \mathbb{P}^2(\mathbb{C}) \mid x^n + y^n = z^n \right\}.$$

(It is sometimes more useful, because of the greater symmetry, to define F_n by the equation $x^n + y^n + z^n = 0$, obtained by replacing z with ζz where $\zeta^n = -1$.) By taking $z = 1, y = 1$ and $x = 1$ respectively, we see that F_n is covered by three copies of the affine curve F_n^{aff}, which omit the points of F_n where z, y and x are zero. For the first affine curve we can take chart maps of the form $[x, y, z] \mapsto x/z$ or y/z, and similarly for the other two. This is a typical example of a smooth projective algebraic curve. Of course, because $\mathbb{P}^2(\mathbb{C})$ is compact, its closed subset F_n is also compact, whereas affine Fermat curves are not.

The great advantage of using projective algebraic curves is that they are compact. There are a number of very useful theorems about Riemann surfaces which have compactness among their hypotheses; these include various numerical results such as the **Riemann-Hurwitz formula**, which we shall state shortly. However, there is a disadvantage in passing from an affine model of a Riemann surface to a projective model: we need an extra coordinate (generally three, rather than

two, when we use plane models), and each point no longer determines its coordinates uniquely, but rather their ratios, which can be less convenient (in defining chart maps, for instance). A good compromise is to represent a projective algebraic curve as the union of two or more affine curves, and then to work with whichever model is most convenient.

Example 6. *Let us try compactify the hyperelliptic curve X^{aff}, given by*

$$y^2 = \prod_{j=1}^{n} (x - a_j)\,,$$

which we considered in previous examples. The corresponding projective curve X^{proj} is given by the equation

$$y^2 z^{n-2} = \prod_{j=1}^{n} (x - a_j z)\,.$$

As in the case of the Fermat curves, this is compact. If $z \neq 0$ then since these are projective coordinates we can take $z = 1$, giving the original affine curve $X^{aff} \subset X^{proj}$; on the other hand, if $z = 0$ then $x^n = 0$ (provided $n \geq 3$) and so $x = 0$, giving a single point $p_\infty = [0, 1, 0]$ as the complement of X^{aff} in X^{proj}. We can use either x or y as a local coordinate on X^{aff}, as we are away from or near a point $p_j = [a_j, 0, 1]$.
Similarly, if $y \neq 0$ we can take $y = 1$, giving an affine curve $Y^{aff} \subset X^{proj}$ with equation $z^{n-2} = \prod_{j=1}^{n} (x - a_j z)$;. its complement consists of the n points p_j, so X^{proj} is the union of these two affine curves.
In order to define local coordinates near p_∞, which is in Y^{aff} but not in X^{aff}, we would like to apply the implicit function theorem to the polynomial $h(x, z) = z^{n-2} - \prod_{j=1}^{n} (x - a_j z)$. Unfortunately, logarithmic differentiation shows that if $n > 3$ then $\partial h/\partial x = \partial h/\partial z = 0$ at p_∞, so the theorem does not apply. Instead, let us go back to X^{aff}, and write its defining equation as $y^2 = q(x) := \prod_{j=1}^{n} (x - a_j)$.
We may assume that each $a_j \neq 0$, by replacing x with $x - a$ for some constant a if necessary. Let us define new variables s and t by

$$t := \frac{1}{x} \quad \text{and} \quad s := \frac{y}{x^{g+1}}$$

where $g := \lfloor (n-1)/2 \rfloor$, so that

$$\deg q = n = \begin{cases} 2g + 1 & (n \text{ odd }) \\ 2g + 2 & (n \text{ even }) \end{cases}$$

The equation $y^2 = q(x)$ is then equivalent to $s^2 = k(t)$ at all points with $x \neq 0$, where

$$k(t) := t^{2g+2} q\left(\frac{1}{t}\right) = \frac{q(x)}{x^{2g+2}}$$

is a polynomial of degree $2g+2$ in $\mathbb{C}[t]$ with simple zeros at the points $t = a_j^{-1}$, and also at $t = 0$ if n is odd. Note that

$$x = \infty \Leftrightarrow t = 0 \Rightarrow s = \sqrt{k(0)} = \begin{cases} 0 & (n\,odd), \\ \pm 1 & (n\,even). \end{cases}$$

We can now apply the implicit function theorem to the polynomial $f^*(s,t) = s^2 - k(t)$ since, as in previous examples, its two partial derivatives do not simultaneously vanish. Specifically, if n is even then we can use t as a local coordinate since $\partial f^*/\partial s = 2s \neq 0$, while if n is odd, so that $\partial f^*/\partial s = 2s = 0$, we can use s since $\partial f^*/\partial t \neq 0$.

This calculation illustrates an important general point about the Riemann surface X associated with the hyperelliptic curve $y^2 = q(x)$. The projection $(x,y) \mapsto x$ realises it as a 2-sheeted covering of $\hat{\mathbb{C}}$, branched over the roots a_j of q, and also over ∞ if n is odd: in the latter case there is a single point $(s,t) = (0,0)$ where $x = \infty$, whereas if n is even there are two points $(s,t) = (\pm 1, 0)$. This can be explained by writing y as a 2-valued function

$$y = \sqrt{(x-a_1)\ldots(x-a_n)},$$

so that each point $x \neq a_j, \infty$ in $\hat{\mathbb{C}}$ is covered by two points $(x, \pm y)$. Now let $x = re^{i\theta}$ for fixed $r > |a_1|, \ldots, |a_n|$, and let θ increase by 2π, so that x follows a circular path enclosing all the roots of q, or equivalently, enclosing the point $\infty \in \hat{\mathbb{C}}$; each factor $\sqrt{(x-a_j)}$ is multiplied by $e^{i\pi} = -1$, so y changes sign, that is, (x,y) passes from one sheet to the other, if and only if n is odd. (A similar argument explains the branching at roots a_j of q for all n : if x follows a small closed path enclosing a_j but no other roots, then just one factor $\sqrt{(x-a_j)}$ is multiplied by -1 , while the rest are unchanged.)

The projective model X^{proj} of X discussed earlier obscures this distinction between odd and even values of n : in either case, it has a single point $p_\infty = [0,1,0]$ at infinity. When n is odd this is where the covering is branched, but when n is even it represents a singularity in the model, where two sheets of the Riemann surface X intersect.

We can use the 2 -sheeted covering $X \to \hat{\mathbb{C}}$ to construct a topological model of X by taking two copies of $\hat{\mathbb{C}}$, one for each branch of $\sqrt{q(x)}$, and joining them across disjoint cuts between $g+1$ pairs of branch-points, namely the roots of q if $n = 2g+2$ is even, together with ∞ if $n = 2g+1$ is odd. A topological model of X^{proj} is then formed by identifying the two points $(s,t) = (\pm 1, 0)$

of X over ∞ if n is even.

Now, we just mention some basic facts on the topology of Riemann surfaces that we will use in the next sections:

1. Every Riemann surface is orientable! This is because the transition functions are biholomorphic, and therefore preserve the orientation.

2. Riemann surfaces can also be triangulated. In fact, this is true more generally for all topological surfaces. This may seem intuitively obvious, but in fact the proof, by Radó in 1925, is not straightforward. (The corresponding result for three dimensional manifolds is also true, but it is false in dimension 4, Freedman's E_8 manifold providing a counterexample). However, there is a simpler proof for compact Riemann surface X if one accepts the existence of a non-constant (and hence surjective) function $f : X \to \hat{\mathbb{C}}$: construct a triangulation $\mathcal{T}$ of $\hat{\mathbb{C}}$ such that the vertices include the critical values, that is, the images of all the ramification points of f, and each face is sufficiently small that f is injective on each connected component of its inverse image; then $f^{-1}(\mathcal{T})$ is a triangulation of X.

3. For compact orientable surfaces X (including compact Riemann surfaces), the genus $g(X)$ is a more commonly-used invariant than the Euler characteristic; this is defined by

$$2 - 2g(X) = \chi(X)$$

   In topology, one shows that such a surface X is homeomorphic to a sphere with $g(X)$ handles attached. Thus $g(X)$ is a non-negative integer, so that $\chi(X)$ is an even integer, with $\chi(X) \leq 2$.

4. (Riemann-Hurwitz Formula) If $f : X \to Y$ is a non-constant holomorphic mapping of compact Riemann surfaces, then

$$2g(X) - 2 = (\deg f)(2g(Y) - 2) + \sum_{p \in X} (\mathrm{mult}_p f - 1)$$

   (Note that the sum on the right-hand side is finite, since $\mathrm{mult}_p f = 1$ for all but finitely many points $p \in X$.)

5. (Riemann Mapping Theorem) If Y is a simply connected Riemann surface, then Y is isomorphic to $\hat{\mathbb{C}}$ or $\mathbb{C}$ or $\mathbb{H}$ (which is isomorphic to $\mathbb{D}$).

Some important application of these facts are given in the next proposition:

Proposition 3. *1. We have $g(Y) \leq g(X)$, with equality if and only if either f is an isomorphism (unramified), or $g(X) = g(Y) = 1$ and f is unramified. If $g(X) > g(Y) = 0$ or 1 , then f is ramified.*

2. *The Fermat curve F_n has genus $(n-1)(n-2)/2$. This can be seen by considering the function $f : F_n \to \hat{\mathbb{C}}, [x, y, z] \mapsto x/z$. On the affine part of the curve given by $z = 1$ and $x^n + y^n = 1$, we have $f : (x, y) \mapsto x$. This shows that $\deg f = n$, since for general x there are n solutions $y \in \mathbb{C}$ of the equation $x^n + y^n = 1$. The exceptions are those points x with $x^n = 1$, where f has only one pre-image, giving us n points $p = (x, 0)$ with mult $_p f = n$. The points of F_n on t he line at infinity $z = 0$ are those of the form $[\zeta, 1, 0]$ where $\zeta^n = -1$, giving n simple (therefore unramified) poles of f. The Riemann-Hurwitz formula now implies that $2g(F_n) - 2 = n(-2) + n(n-1) = n^2 - 3n$, and so:*

$$g(F_n) = \frac{(n-1)(n-2)}{2}.$$

3. *More generally, if $f : X \to \hat{\mathbb{C}}$ is a non-constant meromorphic function on a compact Riemann surface X, then*

$$g(X) = 1 - \deg f + \frac{1}{2} \sum_{p \in X} (\mathrm{mult}_p f - 1).$$

4. *The genus of the hyperelliptic Riemann surface X considered in examples of this section is just the integer g given by the example 6.*

5. *(Considering theory of covering spaces) Every Riemann surface X is isomorphic to the quotient space $\Gamma \backslash Y$ where Y is the universal covering space of X and Γ is the* ***covering group****, the group consisting of all* ***covering transformations****, those $\sigma \in \mathrm{Aut}\, Y$ with $F \circ \sigma = F$; this group permutes the fibres of F transitively.*

One important consequence of uniformization is on the calculation of automorphism group of compact Riemann surfaces. It will directly lead to different kind of geometries like hyperbolic geometry on complex surfaces. We just mention the following theorem as an important result in this way:

Proposition 4. 1. *If $Y = \hat{\mathbb{C}}$, then $\mathrm{Aut}\, \hat{\mathbb{C}} = \mathrm{PSL}_2(\mathbb{C})$. The only covering group acting on Y is $\Gamma = \{\mathrm{id}\}$, so $X = \Gamma \backslash Y = \hat{\mathbb{C}}$. The genus of X is 0.*

2. *If $Y = \mathbb{C}$ then $\mathrm{Aut}\, \mathbb{C} = \mathrm{AGL}_1(\mathbb{C}) = \{z \mapsto az + b \mid a \in \mathbb{C}^*, b \in \mathbb{C}\}$. The covering group Γ consists of translations and is either trivial, infinite cyclic, or a lattice Λ, so that $X \cong \mathbb{C}$ or $\mathbb{Z} \backslash \mathbb{C}$ or a torus (elliptic curve) $\Lambda \backslash \mathbb{C}$, respectively. For example in the case $X = \mathbb{Z} \backslash \mathbb{C}$ we have*

$$F(z) = \exp(z), \quad F : \mathbb{C} \to \mathbb{C}^* = \mathbb{C} \backslash \{0\}$$

and $\exp(z + 2\pi i k) = \exp(z)$ for all $k \in \mathbb{Z}$. The only compact quotients are the tori, which have genus $g = 1$.

3. *We have $Y = \mathbb{H} \cong \mathbb{D}$ in all other cases, and in particular for compact Riemann surfaces X if and only if they have genus $g > 1$. Here Γ is a subgroup of Aut $\mathbb{H} = \mathrm{PSL}_2(\mathbb{R})$ and is called a (torsion-free)* ***Fuchsian group****.*

We finish this section with a brief explanation on algebraic curves and elliptic curves. We use the term of algebraic curves, as it is usually defined in the context of basic algebraic geometry. The most important result for our purpose is the following:

Proposition 5. *There is an equivalence between the two categories of compact Riemann surfaces and of smooth complex projective algebraic curves.*

An equally important result for us is Belyi's theorem, which we will also discuss in detail very soon. This shows that within these two categories, the compact Riemann surfaces which are obtained from special maps (that is, embedded graphs) correspond to the smooth projective algebraic curves which are defined over algebraic number fields (more precisely: those defined in suitable coordinates by equations in which the coefficients are algebraic numbers). The automorphisms of these algebraic number fields, and their effect on the corresponding Riemann surfaces and maps, lead us naturally into a kind of visualization of Galois groups. We will see this idea in the next sections.

Let us explain two different (but equivalent!) approaches to elliptic curves (also look at proposition 4):

Let ω_1 and ω_2 be elements of $\mathbb{C}$ which are linearly independent over $\mathbb{R}$. The additive subgroup

$$\Lambda = \Lambda(\omega_1, \omega_2) = \{m\omega_1 + n\omega_2 \mid m, n \in \mathbb{Z}\}$$

which they generate is called a lattice, and we call ω_1 and ω_2 a basis for Λ. This subgroup Λ is discrete, meaning that every $\omega \in \Lambda$ has an open neighbourhood in $\mathbb{C}$ containing no other elements of Λ.

Given a lattice Λ, there is an equivalence relation on $\mathbb{C}$ given by $z_1 \equiv z_2 \bmod \Lambda$ if and only if $z_1 - z_2 \in \Lambda$: the equivalence classes are the cosets $z + \Lambda$ of Λ in $\mathbb{C}$, and the quotient space is denoted by $\mathbb{C}/\Lambda$. Since the group $\mathbb{C}$ is abelian, Λ is a normal subgroup and hence $\mathbb{C}/\Lambda$ is a group, with its structure inherited from $\mathbb{C}$.

The parallelogram $P = \{x\omega_1 + y\omega_2 \mid x, y \in [0, 1]\}$ is a **fundamental region** for Λ, that is, each $z \in \mathbb{C}$ is equivalent to an element of P, and if two elements of P are equivalent, then they lie on the boundary ∂P of P. Since each coset of Λ has a representative in P, we can form $\mathbb{C}/\Lambda$ by identifying equivalent points $z_1, z_2 \in \partial P$, specifically by identifying pairs of opposite sides $x = 0, 1$ and $y = 0, 1$ of P to form a torus. The holomorphic structure on $\mathbb{C}$ yields a holomorphic structure on $\mathbb{C}/\Lambda$, making it a Riemann surface. This surface is compact, since it is a continuous image of P, which is compact by the Heine-Borel theorem. It is an easy exercise to find a triangulation of

$\mathbb{C}/\Lambda$ which shows that this surface has genus 1.
It is also possible to obtain this Riemann surface $\mathbb{C}/\Lambda$ from an elliptic curve. To establish the link between these two approaches, we need to use elliptic functions. These are meromorphic functions on $\mathbb{C}$ which are doubly periodic with respect to some lattice Λ. To say that f is doubly periodic with respect to Λ means that

$$f(z+\omega) = f(z) \quad \text{for all } z \in \mathbb{C} \text{ and all } \omega \in \Lambda.$$

For a given lattice Λ, the elliptic functions with respect to Λ form a field $F(\Lambda)$ under the obvious operations, such as $(f+g)(z) = f(z)+g(z)$. We can think of these functions as the meromorphic functions on $\mathbb{C}/\Lambda$ by defining $f(z+\Lambda) = f(z)$ (welldefined by double periodicity). Since $\mathbb{C}/\Lambda$ is compact, the theory of meromorphic functions works nicely for this Riemann surface.
The constant functions are obvious examples of elliptic function, but we also need some non-constant examples. **The Weierstrass function** is defined to be

$$\wp(z) = \wp_\Lambda(z) = \frac{1}{z^2} + \sum_\omega{}' \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right)$$

where $\sum'$ denotes the sum over all $\omega \neq 0$ in Λ. This series is uniformly convergent on compact subsets of $\mathbb{C}\backslash\Lambda$, so $\wp$ is meromorphic on $\mathbb{C}$, with poles of order 2 at the lattice points. To show that $\wp$ is double periodic, first consider

$$\wp'(z) = -2 \sum_\omega \frac{1}{(z-\omega)^3}$$

This is meromorphic, with poles of order 3 at the lattice-points. It can be Shown that $\wp'$ is doubly periodic with respect to Λ and also that $\wp$ is doubly periodic.
Thus $\wp, \wp' \in F(\Lambda)$ so the field $\mathbb{C}(\wp, \wp')$ of rational functions of $\wp$ and $\wp'$ is contained in $F(\Lambda)$. In fact, $F(\Lambda) = \mathbb{C}(\wp, \wp')$, so that every elliptic function can be expressed as a rational function of $\wp$ and $\wp'$. The functions $\wp$ and $\wp'$ are not algebraically independent: by comparison of the Laurent series on both sides, it is not hard to prove that they satisfy a differential equation

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3$$

where $g_2 = g_2(\Lambda) = 60G_4$ and $g_3 = g_3(\Lambda) = 140G_6$, and where $G_k = G_k(\Lambda) = \sum_\omega' \omega^{-k}$ for $k = 4, 6$ (these are called **the Eisenstein series for** Λ). Now compare this equation with **the Weierstrass normal form**

$$y^2 = 4x^3 - c_2x - c_3$$

for an elliptic curve E. Given any $c_2, c_3 \in \mathbb{C}$ such that the discriminant $\Delta := c_2^3 - 27c_3^2$ is non-zero,

one can show that there is a lattice Λ in $\mathbb{C}$ with $g_2(\Lambda) = c_2$ and $g_3(\Lambda) = c_3$. We can then write $x = \wp(z)$ and $y = \wp'(z)$, where $\wp = \wp_\Lambda$ for this lattice Λ. This allows us to identify each point $(x, y) \in E$ with the corresponding point $z + \Lambda \in \mathbb{C}/\Lambda$, so we identify the affine elliptic curve E with $\mathbb{C}/\Lambda$ minus the point $0+\Lambda$, where $\wp$ and $\wp'$ have poles. Embedding E in its projective closure $E^{\text{proj}} = E \cup \{p_\infty\}$, we can extend this identification to the entire torus $\mathbb{C}/\Lambda$ by identifying the point $p_\infty = [0, 1, 0]$ at infinity in E^{proj}, with $0 + \Lambda$. This identification is in fact a biholomorphic map, so we get the following theorem:

Theorem 1. *Every torus is isomorphic to a smooth projective algebraic curve whose affine part can be described by a Weierstrass equation.*

2.2 Galois Theory

We will review quickly some facts on finite dimension Galois theory and then we will discuss the infinite dimensional Galois theory, since $Gal(\bar{\mathbb{Q}}/\mathbb{Q})$ is one the most importantn objects in this note (and maybe in the whole of mathematics!). In the next section, which is mostly descriptive and informal, we will explain a similarity between two theorems. One is the **Galois correspondance** and the other one is **classification of connected coverings** in algebraic topology. This similarity is actually coming from the notion of **Galois categories** which will be discussed in a seperate and new note.

Let's start by reviewing basic definitions.
For an extension of fields L/K, write $\mathrm{Aut}(L/K)$ for the K-automorphisms of L : these are the isomorphisms $L \to L$ that fix each element of K. When L/K is a finite extension:
(1) every $\alpha \in L$ is algebraic over $K : f(\alpha) = 0$ for some nonzero $f(X) \in K[X]$,
(2) the group $\mathrm{Aut}(L/K)$ is finite with size $\leq [L : K]$.

Example 7. *1.* $\mathrm{Aut}(\mathbb{Q}(i)/\mathbb{Q}) = \{\alpha \mapsto \alpha, \alpha \mapsto \bar{\alpha}\}$.
2. $\mathrm{Aut}(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q})$ *is trivial, even though* $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$.
3. Aut $(\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q})$ *has size 2 even though* $[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 4$ *: the automorphisms send* $\sqrt[4]{2}$ *to* $\pm\sqrt[4]{2}$ *and both fix* $\sqrt{2}$*, so* $\mathrm{Aut}(\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}) = \mathrm{Aut}(\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2}))$.

The following proposition, is what we need to define Galois extensions:

Proposition 6. *The following properties turn out to be equivalent for a finite extension* L/K *:*
(1) $|\mathrm{Aut}(L/K)| = [L : K]$,
(2) L *is a splitting field over* K *of a separable polynomial in* $K[X]$,
(3) The only elements of L *fixed by* $\mathrm{Aut}(L/K)$ *are the elements of* K,
(4) L/K *is both separable (every element of* L *has a separable minimal polynomial over* K *) and normal (every irreducible polynomial in* $K[X]$ *that has a root in* L *splits completely over* L *).*

Definition 2. *when these properties hold, L/K is called a Galois extension and we write* $\mathrm{Aut}(L/K)$ *as* $\mathrm{Gal}(L/K)$*, calling it the Galois group of L over K.*

Let's see an example (which is not so straightforward!):

Example 8. *For an odd prime p, let $L = \mathbb{Q}\left(\sqrt[p]{2}, \zeta_p\right)$ where ζ_p is a nontrivial pth root of unity. This is the splitting field over $\mathbb{Q}$ of $X^p - 2$, which is irreducible (and separable, as we're in characteristic 0). The elements σ of* $\mathrm{Gal}(L/\mathbb{Q})$ *are determined by the values of $\sigma\left(\zeta_p\right)$ and $\sigma(\sqrt[p]{2})$:*

$$\sigma\left(\zeta_p\right) = \zeta_p^{a(\sigma)}, \quad \sigma(\sqrt[p]{2}) = \zeta_p^{b(\sigma)} \sqrt[p]{2}$$

where $a(\sigma) \in (\mathbb{Z}/p\mathbb{Z})^\times$ and $b(\sigma) \in \mathbb{Z}/p\mathbb{Z}$.

Composition of two automorphisms σ and τ in $\mathrm{Gal}(L/\mathbb{Q})$ *affects the exponents on ζ_p in the same way that the matrices*

$$\begin{pmatrix} a(\sigma) & b(\sigma) \\ 0 & 1 \end{pmatrix}$$

multiply:

$$\sigma\left(\tau\left(\zeta_p\right)\right) = \sigma\left(\zeta_p^{a(\tau)}\right) = \sigma\left(\zeta_p\right)^{a(\tau)} = \left(\zeta_p^{a(\sigma)}\right)^{a(\tau)} = \zeta_p^{a(\sigma)a(\tau)}$$

and

$$\sigma(\tau(\sqrt[p]{2})) = \sigma\left(\zeta_p^{b(\tau)} \sqrt[p]{2}\right) = \sigma\left(\zeta_p\right)^{b(\tau)} \sigma(\sqrt[p]{2}) = \left(\zeta_p^{a(\sigma)}\right)^{b(\tau)} \zeta_p^{b(\sigma)} \sqrt[p]{2} = \zeta_p^{a(\sigma)b(\tau)+b(\sigma)} \sqrt[p]{2}.$$

This matches the equation

$$\begin{pmatrix} a(\sigma) & b(\sigma) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a(\tau) & b(\tau) \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a(\sigma)a(\tau) & a(\sigma)b(\tau) + b(\sigma) \\ 0 & 1 \end{pmatrix}.$$

Therefore $\mathrm{Gal}(L/\mathbb{Q})$ *is isomorphic to the matrix group* $\left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a \in (\mathbb{Z}/p\mathbb{Z})^\times, b \in \mathbb{Z}/p\mathbb{Z} \right\}$.

For a finite Galois extension L/K, the Galois correspondence associates to each subgroup H of $\mathrm{Gal}(L/K)$ the intermediate field $L^H = \{\alpha \in L : \sigma(\alpha) = \alpha \text{ for all } \sigma \in H\}$, where $K \subset L^H \subset L$, and to each intermediate field E the subgroup $\mathrm{Gal}(L/E)$, which fixes E. This is illustrated by the following field diagram:

$$\begin{tikzcd}
L \arrow[r, leftrightarrow] \arrow[d, dash] & \{id\} \arrow[d, dash] \\
E \arrow[r, leftrightarrow] \arrow[d, dash] & H \arrow[d, dash] \\
K \arrow[r, leftrightarrow] & G
\end{tikzcd}$$

$$H = Gal(L/E), E = L^H = \{\alpha \in L | \forall \sigma \in H : \sigma(\alpha) = \alpha\}$$

The main theorem of the Galois theory, is the following one. This is the theorem which we will refer to it many times in the rest of this section:

Theorem 2. *(**Galois Correspondence**). Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$. The inclusion-reversing mappings $E \mapsto \mathrm{Gal}(L/E)$ and $H \mapsto L^H$ between the intermediate fields between K and L and the subgroups of G are inverses of each other and satisfy the following properties when E and H correspond $\big(E = L^H, H = \mathrm{Gal}(L/E)\big)$:*
(a) $|H| = [L : E]$ and $[E : K] = [G : H]$,
(b) two intermediate fields E and E', with corresponding subgroups H and H', are isomorphic over K if and only if H and H' are conjugate subgroups of G; in particular, $\mathrm{Gal}(L/\sigma(E)) = \sigma\,\mathrm{Gal}(L/E)\sigma^{-1}$ for $\sigma \in G$,
(c) E/K is Galois if and only if $H \triangleleft G$, in which case the restriction map $G \to \mathrm{Gal}(E/K)$, where $\sigma \mapsto \sigma|_E$, is surjective with kernel H, so $G/H \cong \mathrm{Gal}(E/K)$.

But there are extensions which are not finite.

Example 9. *Three algebraic extensions of $\mathbb{Q}$ that are not finite extensions are the composite of all quadratic fields*

$$\mathbb{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \ldots),$$

the p-power cyclotomic extension

$$\mathbb{Q}\left(\zeta_{p^\infty}\right) = \bigcup_{n \geq 1} \mathbb{Q}\left(\zeta_{p^n}\right)$$

where $\zeta_{p^n} = e^{2\pi i/p^n}$, and the algebraic closure $\overline{\mathbb{Q}}$.

Even when an algebraic extension L/K has infinite degree, it has a built-in finiteness: every finite set of elements of L lies in a finite subextension of K. That is, every algebraic extension is a union of finite extensions.
The following theorem is the same as proposition 5 and we use it to define Galois extension, when the degree of extension is not finite.

Theorem 3. *For an algebraic extension L/K, the following properties are equivalent:*
(1) $L = \bigcup_i L_i$, with each L_i/K a finite Galois extension,
(2) L is the splitting field over K of a set of separable polynomials in $K[X]$,
(3) $L^{\mathrm{Aut}(L/K)} = K$,
(4) L/K is both separable (every element of L has a separable minimal polynomial over K) and normal (every irreducible polynomial in $K[X]$ that has a root in L splits completely over L).

Definition 3. *We call an algebraic extension L/K Galois if it satisfies the conditions in the previous theorem. When L/K is a Galois extension, we set its Galois group $Gal(L/K)$ to be the group of all K-automorphisms of L.*

Note that all the examples in example 9 are Galois.
When L/K is an infinite Galois extension, it is often impossible to write down concrete formulas for elements of $\mathrm{Gal}(L/K)$. What we can do is indicate how we want an automorphism to look on a subfield and then know it can be extended (in many ways) to an automorphism in $\mathrm{Gal}(L/K)$ by using Zorn's lemma.

Theorem 4. *Let L/K be a Galois extension:*
(1) Every K-isomorphism $\varphi : E \to E'$ between two intermediate fields between K and L extends to a K-automorphism $\sigma : L \to L$.
(2) If E/K is Galois then the restriction homomorphism $\mathrm{Gal}(L/K) \to \mathrm{Gal}(E/K)$ where $\sigma \mapsto \sigma|_E$ is surjective.

For instance, the conjugation automorphism in $\mathrm{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$ where $a + b\sqrt{2} \mapsto a - b\sqrt{2}$ can be extended (or "lifted") all the way up to an element of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, but it's hopeless to expect any kind of general formula for elements of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ other than the identity and complex conjugation.
The following two examples are about infinite-degree Galois extensions L/K where all the elements of $\mathrm{Gal}(L/K)$ have concrete descriptions:

Example 10. *The group $\mathrm{Gal}(\mathbb{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \ldots)/\mathbb{Q})$ is the countable direct product (not direct sum!) of copies of $\{\pm 1\}$, each factor being a choice of sign by which an automorphism affects $\sqrt{-1}$ or $\sqrt{p}$ for a prime p.*

The following example is an important one which is worth mentioning:

Example 11. *For prime p, since $\mathbb{Q}(\zeta_{p^\infty}) = \bigcup_{n\geq 1} \mathbb{Q}(\zeta_{p^n})$ we can describe an element σ of $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q})$ by indicating what σ looks like on each field $\mathbb{Q}(\zeta_{p^n})$. By finite Galois theory, $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^n})/\mathbb{Q}) \cong (\mathbb{Z}/p^r\mathbb{Z})^\times$ by $\sigma(\zeta_{p^n}) = \zeta_{p^n}^{a_n}$ for some integer $a_n \bmod p^n$ where $(a_n, p) = 1$.*

$$\begin{tikzcd}
\mathbb{Q}(\zeta_{p^\infty}) \arrow[d, dotted, no head] \\
\mathbb{Q}(\zeta_{p^3}) \arrow[d, no head] \\
\mathbb{Q}(\zeta_{p^2}) \arrow[d, no head] \\
\mathbb{Q}(\zeta_p)
\end{tikzcd}$$

Each σ gives us a list of numbers $a_n \bmod p^n$ in $(\mathbb{Z}/p^n\mathbb{Z})^\times$, but they are not independent of each other: there is a compatibility condition between them: from $\zeta_{p^{n+1}}^p = \zeta_{p^n}$ we have

$$\sigma\left(\zeta_{p^{n+1}}^p\right) = \sigma\left(\zeta_{p^n}\right) \Longrightarrow \sigma\left(\zeta_{p^{n+1}}\right)^p = \zeta_{p^n}^{a_n} \Longrightarrow \left(\zeta_{p^{n+1}}^{a_{n+1}}\right)^p = \zeta_{p^n}^{a_n} \Longrightarrow \zeta_{p^n}^{a_{n+1}} = \zeta_{p^n}^{a_n}$$

so $a_{n+1} \equiv a_n \bmod p^n$. Two examples of this are (i) $a_n = a$ for a common integer a and (ii) $a_n = 1 + p + \cdots + p^{n-1}$.

Conversely, a list of $a_n \bmod p^n \in (\mathbb{Z}/p^n\mathbb{Z})^\times$ for all $n \geq 1$ where $a_{n+1} \equiv a_n \bmod p^n$ for all n leads to an automorphism σ in $\mathrm{Gal}\left(\mathbb{Q}\left(\zeta_{p^\infty}\right)/\mathbb{Q}\right)$.

Finite degree Galois extensions have finite Galois groups. For infinite degree Galois extensions, the Galois group is always infinite:

Theorem 5. *If L/K is an infinite-degree Galois extension then $\mathrm{Gal}(L/K)$ is an infinite group.*

Proof. We'll prove the contrapositive. If $\mathrm{Gal}(L/K)$ is finite, say of order m, then each $\alpha \in L$ has degree at most m over K, so there is a uniform upper bound on the degrees over K of all elements of L. That implies L/K is finite. (see [Lan05]) □

Note that $\mathrm{Gal}(L/K)$ may be an infinite set does not make $\{\sigma(\alpha) : \sigma \in \mathrm{Gal}(L/K)\}$ infinite: that set is always finite since it's the K-conjugates of α; it has enormous repetitions in it. For example if $c \in K$ then $\{\sigma(c) : \sigma \in \mathrm{Gal}(L/K)\} = \{c\}$. An analogue in finite Galois theory is that $\{\sigma(\alpha) : \sigma \in \mathrm{Gal}(L/K)\}$ can have size smaller than $|\mathrm{Gal}(L/K)|$, but in a finite extension there will be some elements (primitive elements) for which $L = K(\alpha)$ and then the number of K-conjugates is $[L : K]$. For infinite Galois extensions that never happens.
The following example explains why there are too many subgroups of these Galois groups to have a one-to-one correspondence between subgroups of the Galois group and intermediate fields:

Example 12. *Let $L = \mathbb{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \ldots)$, so $\mathrm{Gal}(L/\mathbb{Q}) = \prod\{\pm 1\}$, a countable direct product of the group $\{\pm 1\}$. This is an abelian group where each non-identity element has order 2. The group $\mathrm{Gal}(L/\mathbb{Q})$ is uncountable, so $\mathrm{Gal}(L/\mathbb{Q})$ has uncountably many subgroups of order 2 . At the same time, L has only countably many subfields of each 2-power degree over $\mathbb{Q}$. Therefore the subfields of L and the subgroups of $\mathrm{Gal}(L/\mathbb{Q})$ do not have the same cardinality.*

To fix the problem of Galois correspondence for infinite degree Galois extensions, Krull defined a topology on Galois groups so that there is a one-to-one correspondence between intermediate fields and closed subgroups of the Galois group. (For a finite Galois extension this topology turns out to be discrete, which is why it's unnecessary to use topology when learning finite Galois theory). Intuitively, what it means for two automorphisms σ and σ' in $\mathrm{Gal}(L/K)$ to be "close" in this topology is that $\sigma = \sigma'$ on a finite subextension F/K with large degree: the larger the degree, the "closer" the automorphisms because a larger finite extension F/K in L covers more of L.

For now, we will write "E" for an arbitrary intermediate field between L and K ("E" for "Extension of K ") and we will write "F" for a finite intermediate field extension of K inside L ("F" for "finite").

Definition 4. *For $\sigma \in \mathrm{Gal}(L/K)$, a basic open set around σ or a basic open neighborhood of σ is a coset $\sigma\,\mathrm{Gal}(L/F)$ where F/K is a finite extension. A nonempty subset U of $\mathrm{Gal}(L/K)$ is open when each element of U is contained in a basic open set inside of U : for each $\sigma \in U, \sigma\,\mathrm{Gal}(L/F) \subset U$ for some finite extension F/K inside of L.*

Note that every open set around the identity contains $\mathrm{Gal}(L/F)$ for some finite extension F/K in L. Indeed, open sets are defined to be unions of basic open sets and the basic open sets around the identity are defined to be the subgroups $\mathrm{Gal}(L/F)$.
Equivalently, a basic open set of σ in $\mathrm{Gal}(L/K)$ is the set of all elements of $\mathrm{Gal}(L/K)$ that look like σ on some finite extension F/K inside L. The intuition here is that a "small" open set around σ is all the automorphisms in $\mathrm{Gal}(L/K)$ that equal σ on a "big" finite extension of K inside L. Note that $F \subset F' \Rightarrow \mathrm{Gal}\,(L/F') \subset \mathrm{Gal}(L/F)$, so being equal to σ on a bigger subfield corresponds to a smaller open set around σ.

Definition 5. *The open sets in $\mathrm{Gal}(L/K)$ as described above, along with the empty set, define a topology on $\mathrm{Gal}(L/K)$. This topology is called the **Krull topology on Galois groups**.*

Example 13. *For $\mathbb{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \ldots)/\mathbb{Q}$, which has Galois group a countable product of copies of $\{\pm 1\}$, the Krull topology on this product is the product topology where each factor $\{\pm 1\}$ has the discrete topology. Note that the product topology on a direct product of infinitely many discrete spaces that are not 1-point sets is not discrete!*

This topology is a very important (and insightful!) characteristic of Galois group, so let's check its properties:

Theorem 6. *The topology on $\mathrm{Gal}(L/K)$ has the following properties:*

1. *the operations of multiplication*

$$\circ : \mathrm{Gal}(L/K) \times \mathrm{Gal}(L/K) \to \mathrm{Gal}(L/K)$$

   and inversion

$$^{-1} : \mathrm{Gal}(L/K) \to \mathrm{Gal}(L/K)$$

   are continuous.

2. *when $\mathrm{Gal}(L/K)$ is finite, the topology is discrete.*

3. $\mathrm{Gal}(L/K)$ *is Hausdorff*

4. *A subgroup of* $\mathrm{Gal}(L/K)$ *in the Krull topology is open if and only if it is closed with finite index.*

5. *The closed subgroups of* $\mathrm{Gal}(L/K)$ *are* $\mathrm{Gal}(L/E)$ *for intermediate field extensions* E/K *in* L.

6. *The open subgroups of* $\mathrm{Gal}(L/K)$ *are* $\mathrm{Gal}(L/F)$ *for finite extensions* F/K *in* L.

7. *The closed normal subgroups of* $\mathrm{Gal}(L/K)$ *are* $\mathrm{Gal}(L/E)$ *where* E *is a Galois extension of* K *in* L. *Equivalently, a closed subgroup* H *of* $\mathrm{Gal}(L/K)$ *is normal if and only if* L^H/K *is a Galois extension.*

8. *The open normal subgroups of* $\mathrm{Gal}(L/K)$ *are* $\mathrm{Gal}(L/F)$ *where* F *is a finite Galois extension of* K *in* L. *Equivalently, an open subgroup* H *of* $\mathrm{Gal}(L/K)$ *is normal if and only if* L^H/K *is a finite Galois extension.*

9. *The Krull topology on* $\mathrm{Gal}(L/K)$ *is compact.*

10. *The Krull topology on* $\mathrm{Gal}(L/K)$ *is totally disconnected.*

So basically the Galois groups are a kind of topological groups. Abstracting the situation further, a set of characterizing topological features of the Krull topology on a Galois group are that it is (i) compact, (ii) Hausdorff, and (iii) totally disconnected. Topological groups with these properties can be built out of finite groups in a similar way to the identification of $\mathrm{Gal}(L/F)$ with a closed subgroup of the product of finite (Galois) groups. This is essentially the "limit" concept in this category and such groups are called **profinite groups**, a name introduced by J.P.Serre that comes from the longer term **projective limit of finite groups**. (The term "inverse limit" is also used as a synonym for "projective limit".) Profinite groups are an important class of compact topological groups that are quite unlike more classical compact groups studied in analysis (orthogonal groups, unitary groups, etc.). All Galois groups, with their Krull topology, are profinite groups, and just as all finite groups can be shown to arise as a Galois group of some finite Galois extension (with no control over the choice of base field!), all profinite groups arise as the Galois group of some Galois extension. Therefore **profinite groups are precisely the kinds of topological groups occuring as Galois groups**.

We finish this part with the correspondence theorem for infinite dimension Galois extensions:

Theorem 7. *(Krull) Let* L/K *be Galois and set* $G = \mathrm{Gal}(L/K)$, *equipped with the Krull topology. Associate to each intermediate field* E *the subgroup* $\mathrm{Gal}(L/E)$ *of* G *and associate to each subgroup* H *of* G *the intermediate field* $L^H = \{\alpha \in L : h(\alpha) = \alpha \text{ for all } h \in H\}$.
(1) For all $E, \mathrm{Gal}(L/E)$ *is a closed subgroup of* G.
(2) For all $H, \mathrm{Gal}\left(L/L^H\right)$ *is the closure of* H *in* G.

(3) (Galois correspondence) The mappings $E \mapsto \mathrm{Gal}(L/E)$ and $H \mapsto L^H$ are inclusionreversing bijections between the intermediate fields in L/K and the closed subgroups of $\mathrm{Gal}(L/K)$, and they are inverses of each other: $L^{\mathrm{Gal}(L/E)} = E$ and $\mathrm{Gal}\left(L/L^H\right) = H$ when H is closed.
(4) For an arbitrary subgroup $H \subset G, L^H = L^{\bar{H}}$.

2.3 Galois Correspondance in Algebraic Topology

This part is and introduction to some ideas that led to the notion of "Galois Categories" and "Algebraic Fundamental Group of a Scheme" by Grothendieck in [GR71].
Let $(X, \star)$ be any pointed topological space. Recall that the fundamental group of $(X, \star)$ is the group of loops starting and ending at $\star$, up to continuous deformation. It is denoted $\pi_1(X, \star)$. The group structure is given by composition of loops. Also Recall that, given a topological space X, a covering $\tilde{X}$ of X is a topological space $\tilde{X}$, with a map $p : \tilde{X} \to X$, such that for every point $p \in X$, there is a neighborhood U_p of p and a set T (with discrete topology) together with a commutative diagram

$$\begin{tikzcd} f^{-1}(U_p) \arrow[rr] \arrow[dr, "f"'] & & T \times U_p \arrow[dl] \\ & U_p & \end{tikzcd}$$

where the upper map is a homemorphism.
Intuitively, locally around each point p, the inverse image of f are a number of copies of X indexed by the set T. The covering is said to be trivial if we can take $U_p = X$, and to avoid this case will assume Y is connected in what follows.

Example 14. *Consider the circle $S^1 = \{z \in \mathbb{C}, |z| = 1\}$. The map $z \mapsto z^n$ is a covering map of the circle with itself, with the set T being the cyclic group $\mathbb{Z}/n\mathbb{Z}$. Another covering is given by $f : \mathbb{R} \to S^1, f(t) = \exp(2\pi i t)$.*

As the functorial properties of π_1 imply, for a covering map $p : (\tilde{X}, \tilde{\star}) \to (X, \star)$, there is the $p_\star$ associated map between the associated fundamental groups.
The interesting theorem which is a kind of topological dual for theorem 6 is the following (Theorem 1.38 in [Hat05]):

Theorem 8. *Let X be path-connected, locally path-connected, and semilocally simply-connected. Then there is a bijection between the set of basepoint-preserving isomorphism classes o f path-connected covering spaces $p : \left(\tilde{X}, \tilde{x}_0\right) \to (X, x_0)$ and the set of subgroups of $\pi_1(X, x_0)$, obtained by associating the subgroup $p_*\left(\pi_1\left(\tilde{X}, \tilde{x}_0\right)\right)$ to the covering space $\left(\tilde{X}, \tilde{x}_0\right)$. If basepoints are ignored, this correspondence gives a bijection between isomorphism classes of path-connected covering spaces $p : \tilde{X} \to X$ and conjugacy classes of subgroups of $\pi_1(X, x_0)$.*

So the natural question is, where does this similarity comes from. The answer is the following (for the definition of categorical concepts, see [Awo10]):
In the algebraic side (Galois theory of fields) consider the contravariant functor $h^{\circ}_{K_s} = Hom(-, K_s)$ on the category of algebraic extensions of a field K. K_s is the seprable closure of K. The main result is:

Theorem 9. *This functor $h^{\circ}_{K_s}$ induces an equivalence between the category of finite separable extensions of K ($\mathcal{K}_{Fin-Sep}$) and the category of sets endowed with a transitive and continuous action of the absolute Galois group $Gal(K)$.*

In the side of covering spaces, another functor (which is usually called the "Fibre functor") plays the same role and gives the following theorem:

Theorem 10. *Let X be a connected, path-connected, locally path-connected and semilocally simply connected topological space and x_0 a point in X. The category of finite coverings of X ($FinCov_X$) is equivalent to the category of $\widehat{\pi_1(X, x_0)}$-sets, where $\widehat{\pi_1(X, x_0)}$ is the profinite completion of the topological fundamental group. (The notion of G-sets is coming from the standard notion of action of a group G on a category(like Sets).)*

This similarity is rooted in the notion of "Galois Categories". A Galois category $\mathcal{C}$ is a category which is equivalent to a $\pi - FinSet$ where π is a profinite group. Being equivalent to a $\pi - FinSet$ forces $\mathcal{C}$ to have specific characteristics and a special functor $F : \mathcal{C} \to Sets$ exists. Both $\mathcal{K}_{Fin-Sep}$ and $FinCov_X$ are Galois.

3 Belyi's Theorem

In this section we will discuss Belyi's theorem, using concepts from covering spaces in algebraic topology. Let's recall some notions and an important theorem:
Let $\mathbb{P}^1(\mathbb{C})$ be the Riemann sphere. An object of primary interest throughout this chapter will be the Riemann surface obtained by removing the three points $\{0, 1, \infty\}$, that is $\mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}$. Note that for our present purposes, any three distinct points on the sphere would suffice, although later it will be necessary that these three points are actually rational.

Proposition 7. *Let π_1 denote the fundamental group of $\mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}$. π_1 is generated by loops l_0, l_1 and l_∞ around each of the missing points, with $l_0 l_1 l_\infty = 1$ and so it is nothing but F_2, the free group with two generators.*

The following definition is also central in this chapter:

Definition 6. *Consider the pairs (X, f), where X is a compact Riemann surface and the map $f : X \to \mathbb{P}^1(\mathbb{C})$ is a holomorphic map unramified outside $\{0, 1, \infty\}$. For reasons soon to be*

explained, such pairs will be referred to as Belyi pairs and the functions f, as Belyi functions. A morphism from one pair (X, f) to another (X', g) is given by a holomorphic map $\rho : X \to X'$ such that $g \circ \rho = f$.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, dashed, "\rho"'] & \mathbb{P}^1(\mathbb{C}) \\ X' \arrow[ur, "g"'] & \end{tikzcd}$$

We start discussing the importance of Belyi Pairs with the following Theorem:

Theorem 11. *There is a bijection between conjugacy classes of subgroups of finite index in π_1 and isomorphism classes of Belyi pairs (X, f).*

Proof. Firstly, if we consider $\mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}$ merely as a topological surface, then we have the standard bijection between conjugacy classes of subgroups of π_1 and classes of unramified coverings of $\mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}$, where subgroups of finite index correspond to finite coverings.(This is just the classification of covering spaces, see for example [Hat05]). To complete the connection, note that starting with a Belyi pair (X, f), then considering the restriction of f to $Y = f^{-1}\left(\mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}\right)$ and forgetting about the complex structures, gives the required unramified covering. For the converse, we start with an unramified covering $f : Y \to \mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}$. The complex structure on $\mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}$ induces a unique complex structure on Y. There is a unique way up to isomorphism of Y to compactify Y and to extend f to the compactification X to give a branched covering of $\mathbb{P}^1(\mathbb{C})$ unramified outside $\{0, 1, \infty\}$ (This is a standard theorem in Riemann surfaces, see [For12]). This gives the corresponding Belyi pair (X, f), completing the proof. □

The main theorem of this section will be the following which shows importance of Belyi Pairs:

Theorem 12. *(Belyi) Let X be a nonsingular projective algebraic curve defined over $\mathbb{C}$. Then X can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a non-constant morphism named $f : X \to \mathbb{P}^1(\mathbb{C})$ unramified outside $\{0, 1, \infty\}$, i.e. if and only if there exists an f such that (X, f) is a Belyi pair.*

Proof. By a classical result following from Weil's criterion [Wei56], we know that X can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a morphism $f : X \to \mathbb{P}^1(\mathbb{C})$ with all critical values contained in $\overline{\mathbb{Q}}$. Since $\{0, 1, \infty\} \subset \overline{\mathbb{Q}}$, this proves the one direction.
The proof of the other direction is due to Belyi [Bel80] and proceeds along the following lines (following the exposition given by Schneps [Sch93]): Starting with a curve X defined over $\overline{\mathbb{Q}}$, we can find a morphism $f : X \to \mathbb{P}^1(\mathbb{C})$ with all critical values in $\overline{\mathbb{Q}}$ as mentioned above or simply by taking a non-constant element of the function field of X. Now the proof is divided into two steps. In the first we compose f on the left with a polynomial defined over $\mathbb{Q}$ to get a function with all its finite critical values lying in $\mathbb{Q}$ instead of just $\overline{\mathbb{Q}}$. This function is then composed on the left with a rational function defined over $\mathbb{Q}$ such that the resulting function has only two finite critical

values, both in $\mathbb{Q}$, which can be taken to be 0 and 1. This function is then still defined on the original X, thus completing the proof.

In more detail: Consider the set $\mathcal{C}$ of all finite critical values of the original function f as well as their conjugates under the action of $\mathbb{G}$ (this action is explained more in next sections, but in fact it is nothing more than acting the permutations on the coefficients). Let g_0 be a polynomial with rational coefficients vanishing exactly on $\mathcal{C}$ (we can take g_0 to be the product of the minimal polynomials of the elements of $\mathcal{C}$, one for each $\mathbb{G}$-orbit). Now the finite critical values of $g_0 \circ f$ will in general consist of the image of the critical values of f under g_0 (in this case just 0) as well as the critical values of g_0. Then we repeat the process to find a polynomial g_1 vanishing on these critical values and having degree strictly smaller than g_0 (for example by taking g_1 to be the derivative of g_0). Continuing in this manner we find a g_n with degree 0 for some n. The polynomial $h = g_{n-1} \circ g_{n-2} \circ \cdots \circ g_1 \circ g_0 \circ f$ then has all critical values in $\mathbb{P}^1(\mathbb{Q})$.

Now let $\mathcal{C}'$ be the set of finite critical values of h. If $\mathcal{C}'$ is empty we are done. If $\mathcal{C}' = \{\alpha\}$, then composing h with the linear fractional transformation $z \mapsto z - \alpha$ gives the required function. If $\mathcal{C}' = \{\alpha, \beta\}$, then composing h with the linear fractional transformation $z \mapsto \frac{1}{\beta-\alpha} z + \frac{\alpha}{\alpha-\beta}$ maps the ordered triple (α, β, ∞) to the ordered triple $(0, 1, \infty)$, so again we are done. If $\mathcal{C}' = \{\alpha, \beta, \gamma, \ldots\}$, then the previous transformation will take γ to a rational number of the form $\frac{m}{m+n}$. Composing with the rational function

$$z \mapsto \frac{(m+n)^{(m+n)}}{m^m n^n} z^m (1-z)^n$$

takes 0 and 1 to 0 , and $\frac{m}{m+n}$ to 1 , without introducing new ramification values, thereby giving a function with fewer ramification values than the original h. Repeating this process gives the required function after a finite number of steps. $\square$

4 Definition of Dessins

Intuitively, dessins are bipartite graphs drawn on compact surfaces. In this section we will make this definition more precise, and show how the same objects can be described in many different ways.

4.1 Topological Defenition

In this section we will define a dessin as a purely topological structure. To do so, we start with a 2-dimensional, compact, connected, oriented surface X. Note for future reference that these are precisely the types of surfaces that can be equiped with the analytical structure of a compact Riemann surface.

Now we want to give X a 2-dimensional cell complex structure. This involves partitioning X into three parts, the first (X_V) being a finite discrete set of points (0-cells, or vertices), the second

(X_E) homeomorphic to a finite disjoint union of open intervals (1-cells, or edges) and the last is (X_F) homeomorphic to a finite disjoint union of open disks (2-cells, or faces).
Then some consideration will show that the closure of an edge must be homeomorphic to either a circle or a closed interval, and contain either one or two vertices corresponding to these two cases. Recall that a graph is defined by a set of vertices and a set of its subsets, each subset having cardinality one or two. Thus there is a natural way to define a graph G_X associated to X with vertices being the elements of X_V, edges the connected components of X_E and the vertices at the end of an edge being the ones contained in its closure.
G_X is required to be a connected bipartite graph. Being bipartite means that we can partition the set of vertices X_V into two sets such that every edge lies between a vertex from the one set and a vertex from the other set. In other words, there exists a function $\mu : X_V \to \{0, 1\}$ assuming different values on adjacent vertices. This implies for example that G_X has no loops. We will say a vertex is of type 0 or 1.

Definition 7. *A Dessin $\mathcal{C}$ is defined by the data $(X = X_V \cup X_E \cup X_F, \mu)$ satisfying the above properties. G_X is refered to as the associated graph and X the underlying surface. Let $\mathcal{C}'$ be another dessin with data $(X' = X'_V \cup X'_E \cup X'_F, \mu')$. We say $\mathcal{C}$ is isomorphic to $\mathcal{C}'$ if there exists an orientation preserving homeomorphism $\varphi : X \to X'$ which induces homeomorphisms $X_V \cong X'_V, X_E \cong X'_E$ and $X_F \cong X'_F$, and which respects the bipartite structure by satisfying $\mu = \mu' \circ \varphi$ on X_V.*

We will sometimes adopt the convention of speaking of a dessin as a graph, implying of course the associated graph. It is important to note however that although two isomorphic dessins necessarily have isomorphic associated graphs, the converse is not true since the way the graph is embedded into the surface is also important. In fact, as we will see in the next section, a dessin can be completely described be giving its graph and a cyclic ordering of the edges at each vertex corresponding to their ordering on the underlying surface. Figure 1 illustrates some examples of dessins. Notice that the two dessins at the top are isomorphic as graphs but not as dessins.
We will see later that the above description corresponds to taking the pre-image of the unit interval under a function $f : X \to \mathbb{P}^1(\mathbb{C})$. It will sometimes be more convenient however to take the pre-image of the real line, leading to a slightly different description of a dessin as a triangulation of a surface.
More precisely, consider a dessin $\mathcal{C}$ as described above. Enlarge the set of vertices X_V by adding one point in each face. Then extend the set X_E by adding an edge joining each new vertex to each old vertex bordering on the face in which the new vertex was placed. Also extend the function on the vertices to $\mu : X_V \to \{0, 1, \infty\}$ by letting it take the value ∞ on the new vertices.
On the right, figure 2 shows the new vertices as stars and the new edges as dotted lines. Vertices of type 0 and 1 are indicated by white and black circles respectively.
If n was the number of edges of the original dessin, then there will now be $3n$ edges and $2n$ faces.

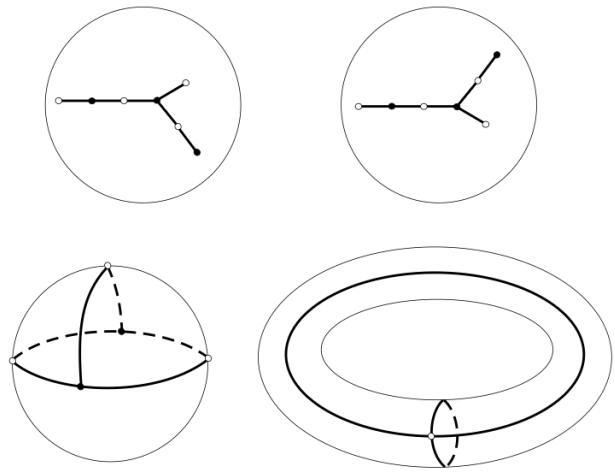


Figure 1: Some examples of dessins

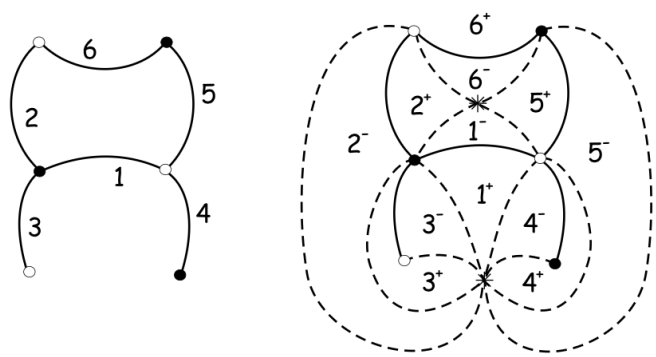


Figure 2: dessins and their triangulation

Each face will be bounded by exactly three edges, hence we have a triangulation of the surface X. The vertices have already been partitioned into three types by means of the values assigned to them by μ. Similarly, we can divide the edges into three types according to the types of vertices linked by the edge, namely $\overline{01}, \overline{0\infty}$ and $\overline{1\infty}$. Finally, the faces, or triangles, can divided into two types according to whether the vertices taken in an anticlockwise direction are arranged $(0, 1, \infty)$, or $(0, \infty, 1)$. We call the triangles positive or negative corresponding to these two cases. Two triangles are said to have the same parity if and only if they are both positive or both negative. Clearly two triangles can only share an edge if they are of different parity.

To go in the reverse direction is now easy: given a triangulation as above, we simply disregard the vertices of type ∞ and the edges of type $\overline{0\infty}$ and $\overline{1\infty}$ to get the original dessin. To summarize, we have shown the following:

Proposition 8. *A dessin can also be described by giving a cell complex $X = X_V \cup X_E \cup X_F$ where*

every face is bounded by exactly three edges, and a function $\mu : X_E \to \{0, 1, \infty\}$ assuming different values on adjacent vertices. An isomorphism is described exactly as before.

We conclude this section with a construction which will be useful in the next section. Consider a usual dessin where the n edges have been labelled from 1 to n. Taking the corresponding triangulation, we have a labelling of the edges of type $\overline{01}$. Since each triangle has exactly one such edge on its boundary, and each such edge is on the boundary of exactly one positive and one negative triangle, we can label the $2n$ triangles as $\{1^+, 1^-, 2^+, 2^-, \ldots, n^+, n^-\}$, where for example triangle $k^+$is the positive triangle having edge k on its boundary. This is illustrated in figure 2.

4.2 Group Theory Definition

All the above information can be codified in a visually less appealing, yet more concise form using groups. Let $F_2 := \langle x, y, z \mid xyz = 1\rangle$ be the free group on two generators x and y. Let F_2 act on the set of edges E of the dessin as follows: x permutes the edges cyclically around each vertex of type 0 in an anticlockwise direction, while y does the same around the vertices of type 1. For example, the permutations induced on the edges of the dessin in figure 2 would be (145)(26)(3) and (123)(4)(56) by x and y respectively.
Since F_2 is acting on a set E with n elements, we get a group homomorphism from F_2 to S_n, the symmetric group on n elements. The image of F_2 under this map is a subgroup of S_n generated by the permutations induced by x and y (denoted by g_0 and g_1 respectively). Note that the action of F_2 is transitive because the graph of the dessin is connected. Then we have the following alternative description of a dessin:

Proposition 9. *There is a bijection between isomorphism classes of dessins and equivalence classes of pairs (g_0, g_1) with $g_0, g_1 \in S_n$ for some n, such that $\langle g_0, g_1\rangle$ is a transitive subgroup of S_n. Two pairs (g_0, g_1) and (g_0', g_1') are considered equivalent if they are simultaneously conjugate in S_n, i.e. if there is some h in S_n such that $g_i' = h^{-1}g_ih$ for $i = 0, 1$.*

Proof. For the one direction it was shown above how to associate a pair (g_0, g_1) with each dessin. In the process it is necessary to label the edges from 1 to n and a different labelling would give another pair which is conjugate to the original one by the permutation h describing the change in labelling.
For the other direction of the proof it is necessary to reconstruct the dessin from a given pair of permutations (g_0, g_1) where $g_0, g_1 \in S_n$. This will be done by joining together $2n$ triangles along their edges to give a triangulation of a surface which describes a dessin as explained in the previous section.
Label the triangles as $1^+, 1^-, 2^+, 2^-, \ldots, n^+, n^-$ and label the three vertices of each as 0,1 and ∞ in such a way that the vertices of the positive (resp. negative) triangles are arranged as described

in the previous section. Use $\left(k^+, \overline{ij}\right)$ to refer to the edge of triangle k^+lying between vertices i and j. Now join the triangles by identifying edges as follows (while orientating the edges to preserve vertex types):

$$\left(k^+, \overline{0\infty}\right) \sim \left(g_0(k)^-, \overline{0\infty}\right)$$
$$\left(k^+, \overline{01}\right) \sim \left(k^-, \overline{01}\right)$$
$$\left(g_1(k)^+, \overline{1\infty}\right) \sim \left(k^-, \overline{1\infty}\right)$$

It is clear from this description that each triangle is joined to exactly three other triangles, all with the opposite parity. Thus we can extend the local orientation of each triangle to a global orientation, so we indeed get a compact, oriented surface. Partitioning the surface into vertices, edges and faces, and defining the natural vertex function μ gives a triangulation description of a dessin as in proposition 2.

It only remains to be checked that the permutations of the edges of dessin correspond to the g_0 and g_1 we started with. This can be verified using the three rules for joining edges given above. Indeed, if we start with an edge of type $\overline{01}$ labelled k and try to determine the cyclic ordering of the edges around its 0-vertex, we find that we need to apply the first two rules alternately, giving the sequence of edges $k, g_0(k), g_0\left(g_0(k)\right), \ldots, k$ as desired. The case for g_1 is similar, completing the proof. □

Given a pair (g_0, g_1), it is possible to deduce many properties of the dessin without having to go through the above process to reconstruct it. If we denote the number of cycles of g_i by $|g_i|$, then $|g_0|$ corresponds to the number of vertices of type 0 and the length of each cycle corresponds to the number of edges emanating from that vertex. A similar statement holds for g_1 and if we define g_∞ as $(g_0 g_1)^{-1}$, then $|g_\infty|$ corresponds to the number of faces of the dessin. The length of each cycle of g_∞ is equal to half the number of edges surrounding the corresponding face. Note that some edges can appear twice on the boundary of a face (for example edges 3 and 4 in figure 2). Passing to the corresponding triangulation, $|g_\infty|$ corresponds to the number of vertices of type ∞. All these properties can be verified for figure 2.

Let $v = |g_0| + |g_1| + |g_\infty|$ denote the total number of vertices of the triangulation, $e = 3n$ the number of edges and $f = 2n$ the number of faces. Then we can calculate the genus g using Euler's formula $2 - 2g = v - e + f$. For example, corresponding to figure 2 we have $g_0 = (145)(26)(3), g_1 = (123)(4)(56)$ and $g_\infty = (1364)(25)$, hence $2-2g = (3+3+2)-3(6)+2(6)$, i.e. $g = 0$, which is what it should be since the underlying surface of this specific dessin is a sphere. This group-theoretic description will be useful later on, but for our immediate purposes we need the following:

Proposition 10. *There is a bijection between isomorphism classes of dessins and conjugacy classes of subgroups of finite index in F_2.*

Proof. To associate a conjugacy class of subgroups of F_2 to a given dessin, consider again the

action of F_2 on E, the set of n edges of dessin, as described in the beginning of this section. This action is considered to be a right action. Now fix a specifix edge e, and let $H \subset F_2$ be the stabilizer of e, i.e. H is the set of all elements in F_2 which act trivially on e. Since F_2 acts transitively on E, H is a subgroup of index n of F_2. This is because two elements of F_2 lie in the same coset of H if and only if they both act on e by taking it to the same edge e'. So the number of cosets of H is equal to the number of ways an element of F_2 can act on e, that is, to the number of edges n. Finally, note that choosing a different edge e gives a conjugate subgroup of F_2, again using the fact that F_2 acts transitively.

For the converse we are given a subgroup H of index n in F_2. Let $\mathcal{C}$ be the set of cosets of H. The group F_2 now acts on $\mathcal{C}$ by taking $a \in F_2$ and $Hb \in \mathcal{C}$ to $Hba \in \mathcal{C}$. This action is clearly transitive, so by denoting the images of the generators x and y of F_2 in $\mathcal{C}$ as g_0 and g_1, we can appeal to proposition 3 to reconstruct a dessin. It only remains to verify that starting with a subgroup of F_2 corresponding to a specific dessin, this procedure will yield the same desssin again, essentially because F_2 acts on the set of cosets in exactly the same way as on the set of edges. More precisely, suppose the H that we started with is the stabilizer of an edge e of some specific dessin. Then we can establish a bijection between the set of edges E of the dessin and the cosets $\mathcal{C}$ by associating an edge e' with the coset Ha where $a \in F_2$ takes e to e' and conversely associating a coset Ha with the image of e in E under the action of a. This bijection commutes with the action of F_2 on E and $\mathcal{C}$ respectively, thus completing the proof. $\square$

5 Grothendieck's Correspondence

The results of the previous sections combine to prove the following theorem, known as the "Grothendieck correspondence":

Theorem 13. *There is a bijection between the isomorphism classes of dessins and the isomorphism classes of Belyi pairs.*

Although we have been working with sets of objects and the concept of an isomorphism between two objects, it would have been possible to define general morphisms between objects and treat the sets of objects as categories. The above correspondence would then have been an equivalence of categories since the bijection respects the morphisms between objects.

As we have seen, the proof of this correspondence takes a detour through group theory. There is however a more explicit way of realizing the correspondence. In the one direction, starting with a Belyi pair (X, f), we can find the corresponding dessin on X simply by taking the preimage of the unit interval under f, i.e. $f^{-1}([0,1])$. The points above 0 will be those of type 0 , and the ones above 1 will be those of type 1 . This construction makes it clear that the number of elements in the fiber above 0 is precisely the number of vertices of type 0 , and order of ramification

at each point in the fiber is precisely the number of edges meeting at the corresponding vertex. Furthermore, the action of F_2 on the edges of the dessin can be seen to correspond directly to the action of the fundamental group π_1 on points of X. More precisely, choose $\frac{1}{2}$ as the base point for the fundamental group and consider the fiber above this point. From the above description, it is clear that there is exactly one point in the fiber on every edge of the dessin. Now the fundamental group acts on this fiber via the lifting of loops. The action of the loop l_0 corresponds exactly to the action of $x \in F_2$ on the edges.

To realize the converse direction more concretely, one must make use of the triangulation associated to a dessin with underlying surface X. Consider the simple triangulation of the sphere by adding the three edges joining the points 0,1 and ∞. There are two triangles, called positive and negative by the original convention. Now, since every triangle is homeomorphic to an open disk, one can define a function f from the triangles of X to the triangles of the sphere, such that f restricted to a single triangle is a homeomorphism between that triangle and the one on the sphere with the same parity. By choosing f appropriately, it is possible to extend it to a continuous function defined on the whole of X, respecting the vertex, edge and face types. This is then the required Belyi function. Using this function we can lift the complex structure on $\mathbb{P}^1(\mathbb{C})$ to a complex structure on X, giving the required Belyi pair.

Note that in the construction just described, one starts with an arbitrary topological surface X, and by drawing a dessin on it, one can equip X with a complex structure. Two questions arise, namely is this complex structure unique, and which complex structures arise in this way? The positive answer to the first question was first pointed out by Grothendieck in [Gro84], but proven essentially before by Jones and Singerman [JS78] and independently by Malgoire and Voisin [VM77] (see also the survey by Wolfart [Wol97]). The answer to the second question is Belyi's theorem, given in the previous section.

Also note that although the Riemann surface X associated to a given dessin is unique, the Belyi function f is only unique up to composing with automorphisms of X. Thus in genus 0 the map f is unique up to $\mathrm{PSL}_2(\mathbb{C})$, in genus 1 up to affine transformation and in genus greater than 1, up to a finite automorphism group. To obtain uniqueness, one needs to restrict f in some way. In genus 0 , this can be done by fixing the position of three seperate points on the dessin since the group $\mathrm{PSL}_2(\mathbb{C})$ acts three-transitively on the sphere.

$$\begin{tikzcd}
Graphs \arrow[rr, leftrightarrow] \arrow[dr, leftrightarrow] & & Algebraic\ Curves \arrow[dl, leftrightarrow] \\
& Compact\ Riemann\ Surfaces &
\end{tikzcd}$$

6 Galois Action on Dessins

6.1 The Action and Faithfullness

An element of $\mathbb{G} = \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ can be viewed as a consistent choice of automorphisms of every finite Galois extension. To make $\mathbb{G}$ into a topological group, start by giving the finite Galois groups the discrete topology. The product of these groups is then given the product topology and $\mathbb{G}$ receives the restriction topology as its subgroup. Equivalently, this is the coarsest topology making all the projection maps from $\mathbb{G}$ to the finite Galois groups, continuous. Consequently, a subgroup of $\mathbb{G}$ is open if and only if it has finite index in $\mathbb{G}$.

Since we can view dessins as Belyi pairs (X, f) with both X and f defined over $\overline{\mathbb{Q}}$, it is natural to consider the action of $\mathbb{G}$ on the dessins. An element $\sigma \in \mathbb{G}$ acts on a pair (X, f) by acting on the coefficients of a defining equation of X and on the coefficients of f. We denote the resulting pair by (X^σ, f^σ).

The question then arises as to whether the action is faithful, that is, whether for any given $\sigma \in \mathbb{G}, \sigma \neq 1$, one can always find a dessin on which σ acts non-trivially. That this is indeed the case follows from the following:

Proposition 11. *The action of $\mathbb{G}$ on the set of genus 1 dessins is faithful.*

Proof. Let σ be an element of $\mathbb{G}$. Choose any algebraic number j on which σ acts non-trivially. Let X be an elliptic curve with j-invariant equal to j. Consider any nonconstant function on X, and using the process described in the proof of Belyi's theorem, compose it with a series of functions to produce a Belyi function f on X. Since X^σ has j-invariant equal to $\sigma(j), \sigma$ acts non-trivially on the pair (X, f). □

The action on the set of genus 0 dessins is also faithful. Indeed, consider the set of genus 0 dessins having only one open cell, that is, only one point above ∞. These dessins are called trees, because by choosing this one point to be ∞, the dessin can be realised as a planar graph with no cycles. Then a result by Lenstra (proved in [Sch94]) states that the action of $\mathbb{G}$ on the set of trees is faithful.

Thus each element of $\mathbb{G}$ can be identified with a unique permutation of the set of dessins, so a better understanding of this action could provide new insight into $\mathbb{G}$. A first step would be to characterize the orbit of a dessin under the group action. By a result of the next section, such an orbit is always finite. However it remains an open problem to decide whether two given dessins lie in the same Galois orbit or not, and some approaches will be described in the next part.

Another consequence of Belyi's theorem is that we can attach a unique number field to each dessin, namely the moduli field of the dessin. This is defined to be the field K of the following lemma (adapted from Wolfart [Wol97]):

Lemma 1. *Let (X, f) be a Belyi pair. Then for a field $K \subset \overline{\mathbb{Q}}$ the following properties are equivalent:*
1. K is the minimal field with the property $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K)$ implies $(X, f) \cong (X^\sigma, f^\sigma)$.
2. For all $\sigma \in \mathbb{G}, \sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K) \Longleftrightarrow (X, f) \cong (X^\sigma, f^\sigma)$.
3. K is the fixed field of $\{\sigma \in \mathbb{G} \mid (X, f) \cong (X^\sigma, f^\sigma)\}$.

We call L **a field of definition of** (X, f) if there is some isomorphic Belyi pair (X', f') with both X' and f' defined over L. Clearly any field of definition contains the moduli field, and in fact the moduli field is the intersection of all possible fields of definition.
One can always find a field of definition which is a number field, hence the moduli field is a number field. From the definition of the moduli field, it follows that the number of elements in the Galois orbit of a given dessin is equal to the degree of its moduli field over $\mathbb{Q}$.

6.2 Galois Invariants

As noted in the previous section, $\mathbb{G} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts faithfully on the set of dessins, dividing it into orbits. Ideally, we would like to have a simple (possibly combinatorial) criterion for telling whether two given dessins lie in the same orbit or not. Currently, the only sure way of doing this is computing a Belyi function for a given dessin and using it to find conjugate dessins. There are however several known necessary (but not sufficient) invariants characterizing the Galois orbit of a dessin. This section introduces the simpler ones.
Just as the Galois action partitions the set of dessins into orbits, the Galois invariants also induce partitions which are by definition coarser. We call an invariant finer in general than another one if it induces a finer partition. Sometimes we will refer to an invariant as being finer in some cases than another simply if it isn't coarser in general. Most invariants seem to be finer than all others in some cases, implying that the best method is a combination of them all.

6.2.1 Elementary Invariants

We make the obvious observation that the bijection $X \leftrightarrow X^\sigma$ performed by a Galois element $\sigma \in \mathbb{G}$ is never holomorphic (not even continuous with the exception of the case in which σ is the complex conjugation). Therefore, X and X^σ will in general be nonisomorphic. Likewise there is not an a priori reason why these two Riemann surfaces should be even homeomorphic. Nevertheless, the action $(X, f)^\sigma = (X^\sigma, f^\sigma)$ preserves the genus as well as some other interesting properties:

Theorem 14. *The action of $\mathbb{G}$ on pairs (X, f) enjoys the following properties:*
(1) $\deg(f^\sigma) = \deg(f)$.
(2) $(f(P))^\sigma = f^\sigma(P^\sigma)$.
(3) $\mathrm{ord}_{P^\sigma}(f^\sigma) = \mathrm{ord}_P(f)$.
(4) The genus of X^σ is the same as the genus of X, i.e. X and X^σ are homeomorphic.

Proof. See [GG12] □

It is also good to think about "What are not invariants" for this action. The simplest answer is the graph itself (and some combinatorial characters of the graph). The examples we illustrate in the section 6 are also useful for this matter.

6.2.2 Valency Type

Consider a dessin $\mathcal{S}$ with n edges.

Definition 8. *The valency of a vertex is the number of edges meeting at the vertex. The valency of a face is half the number of edges bordering on the face, where an edge surrounded by the face is counted twice.*

Since we are dealing with bipartite graphs, the number of edges bordering on a face will always be even, so this definition makes sense. For now we will refer to the faces as vertices of type ∞. Then it is clear from the correspondence between valencies of vertices of type i and orbit lengths of g_i where $i \in \{0, 1, \infty\}$ that the sum of the valencies of vertices of type i equals the number of edges, n.
To make this precise, let m_i be the number of vertices of type $i, i \in \{0, 1, \infty\}$, and label them $1, 2, \ldots, m_i$. Let $k_r^{(i)}$ be the valency of the r 'th vertex of type i. Then for each i there is an unordered partition $n = k_1^{(i)} + k_2^{(i)} + \cdots + k_{m_i}^{(i)}$. We say the dessin has valency type

$$\left(k_1^{(0)} + \cdots + k_{m_0}^{(0)}; k_1^{(1)} + \cdots + k_{m_1}^{(1)}; k_1^{(\infty)} + \cdots + k_{m_\infty}^{(\infty)}\right)$$

For example the valency type corresponding to Figure 2 is $(1 + 2 + 3, 1 + 2 + 3, 2 + 4)$.
It can be shown that the valency type remains invariant under the Galois action (see [Mat06]). Furthermore, there are only a finite number of graphs with a given valency type, hence Galois orbits must be finite. There is no upper bound on their lengths however:

Proposition 12. *Galois orbits of dessins can be arbitrarily large.*

Proof. Let σ be an element of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which is not a torsion element. Then for every dessin s there is a positive number n_s such that $\sigma^{n_s}(s) = s$ (since orbits are finite). Suppose orbits are bounded in length by M, i.e. $n_s \leqslant M$ for all dessins s. Now let $n = M!$. Then $\sigma^n(s) = s$ for all s. But since $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts faithfully, it follows that σ^n must be the identity, i.e. σ is a torsion element. The contradiction shows that orbits are not bounded in length. □

The genus of the underlying surface of the dessin is determined by the valency type. Thus from the invariance of the valency type we can also deduce the invariance of the genus.

6.2.3 Monodromy Groups and Composition

Consider the dessin given by the pair of permutations (g_0, g_1) from S_n. The subgroup of S_n generated by g_0 and g_1 is called the monodromy group of the dessin. This group is another Galois invariant, in general finer than the valency type.
Variations of the monodromy group invariant can be produced by first composing the given dessin with another one and then calculating the monodromy group of the resulting dessin. Given Belyi pairs (X, f) and (Y, g), we would like to define their composition as the Belyi pair $(X, g \circ f)$. Clearly certain restrictions have to be placed on the pair (Y, g) for this to make sense, leading to the definition:

Definition 9. *A Belyi pair (Y, g) is said to be composable if $Y = \mathbb{P}^1(\mathbb{C}), g(\{0, 1, \infty\}) \subset \{0, 1, \infty\}$ and g is defined over $\mathbb{Q}$. Given such a pair and another general pair (X, f), this is a sufficient condition for $(X, g \circ f)$ to be a Belyi pair, called the composition of (X, f) with g.*

Since Y is always $\mathbb{P}^1(\mathbb{C})$, we just refer to g as a composable Belyi function. The associated dessin is called a composable dessin. The reason for requiring g to be defined over $\mathbb{Q}$ is to ensure that composition commutes with the Galois action:

Proposition 13. *Let g be a composable Belyi function. Then if two dessins are in the same Galois orbit, their respective compositions with g will also lie in the same Galois orbit.*

Proof. Let (X, f) be a Belyi pair, and σ an element of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Then the proof of the theorem will follow from the commutativity of the following diagram, where the horizontal arrows indicate composition with g, and the vertical arrows indicate the action of σ.

$$\begin{tikzcd}
(X, f) \arrow[r, "g"] \arrow[d, "\sigma"] & (X, g \circ f) \arrow[d, "\sigma"] \\
(X, f^\sigma) \arrow[r, "g"'] & (X^\sigma, (g \circ f)^\sigma)
\end{tikzcd}$$

Going around clockwise we find $h = (g \circ f)^\sigma$, while going anti-clockwise gives $h = g \circ f^\sigma$. Since g is defined over $\mathbb{Q}$, these functions coincide. $\square$

An examples of composing dessins is shown in figure 3. The dessin in the middle is composed with the dessin drawn next to the arrows, and the resulting composition is labelled A.

6.2.4 Trees

Trees are genus 0 dessins with a single face. They are simpler than general dessins in certain respects, such as always having a polynomial as a Belyi function. Yet as noted before, $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$

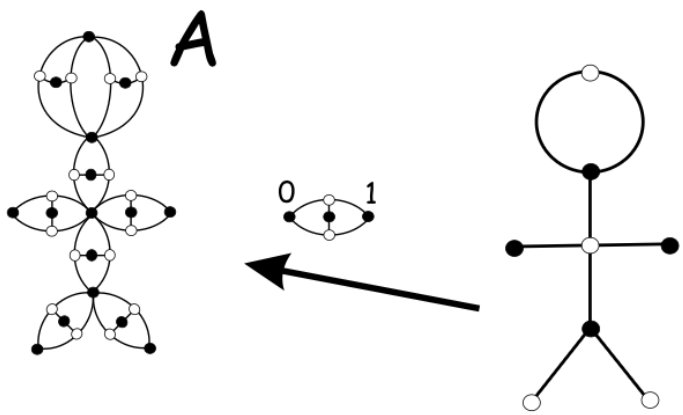


Figure 3: composition of dessins

still acts faithfully on the set of trees. Consequently trees are a natural starting point for studying properties of dessins. In this section we state (a corollary of) a result of [Zap00] which gives an invariant which is strictly finer than valency types for a certain family of trees.

For convenience we change the notation from that given at the beginning of the section. Let T be a specific tree. Let n and m denote the number of vertices of type 1 and 0 respectively. Label the vertices of each type, and let p_i and q_j denote the valencies of the i 'th vertex of type 1 and the j 'th vertex of type 0 respectively. We denote the valency type of the tree by $(p_1, \ldots, p_n; q_1, \ldots, q_m)$. From Euler's formula then follows that $\sum_i p_i = \sum_j q_j = m + n - 1$.

Actually it is necessary to label the vertices of type 1 (the p_i 's) in a specific way to capture the way the graph is embedded into the sphere. First note that this embedding is described by giving a cyclic permutation of the edges meeting at each vertex corresponding to their order in an anticlockwise direction around the vertex. Now choose a vertex of type 1 to be labelled as number 1 and choose an edge leading away from it as the active edge. We now traverse the graph using the following algorithm:

Walk along the active edge to the next vertex. Replace the current active edge by the edge following it in the permutation corresponding to this vertex. Now walk along this edge, repeating the process until reaching the original vertex and same active edge. Along the way, label the vertices of type 1 in the order in which they are first reached. Note that every edge will be traversed exactly twice.

Another way to describe it, is to consider the pair of permutations (g_0, g_1) corresponding to the tree. The order in which the edges are traversed can be found by applying g_0 and g_1 alternately.

Having labelled the type 1 vertices, we now make the assumption that they all have distinct valencies, i.e. that the p_i 's are all distinct. There is then a unique permutation σ on n elements such that $p_{\sigma(1)} < p_{\sigma(2)} < \cdots < p_{\sigma(n)}$. Let $S(T)$ (the signature) be either 1 or -1 corresponding to whether this is an even or odd permutation.

Note that $S(T)$ depends on the choice of starting vertex, in the sense that starting at a different vertex will give a different permutation which could have a different signature. However, in the case where n as well as all the q_i 's are odd (conditions which are met in the following theorem), it can be verified that the signature is independant of the choice of starting vertex.

Theorem 15. *Let $n, m, p_1, \ldots, p_n, q_1, \ldots, q_m$ be positive integers such that:*
1. $p_1 < \cdots < p_n$ and $\sum_i p_i = \sum_j q_j = m + n - 1$.
2. All the q_j 's are odd.
3. $n \equiv 1 (\mathrm{mod} 4)$
4. $p_1 \ldots p_n (p_1 + \cdots + p_n)$ is a perfect square.
Then $S(T)$ is a Galois invariant for the trees T having valency type ($p_1, \ldots, p_n; q_1, \ldots, q_m$). Thus this valency class splits into at least two Galois orbits.

One of the examples which started conjectures in this direction is known as Leila's flowers (after the works of Leila Schneps). They are trees forming a valency class of order 24 which splits into two Galois orbits of equal size. A tree from each orbit is shown in figure ??????????. The vertices of type 1 have distinct valencies $(2, 3, 4, 5, 6)$ and there are $5 (\equiv 1 (\bmod 4))$ of them. All the vertices of type 0 have valency 1 or 5 , hence odd. Finally $p_1 \ldots p_n (p_1 + \cdots + p_n) = 14400 = (120)^2$ is a perfect square. Thus the theorem applies. Labelling the vertices of type 1 as indicated shows that the tree on the right needs the odd permutation $(12)(3)(4)(5)$ to sort the valencies in increasing order, whereas for the tree on the left they are already sorted. Since the identity permutation is even, this shows that they are in different orbits.

7 Explicit Examples

In this section, some examples of Belyi maps and calculations on dessins are discussed. The main sources for this section are [LZ03] and [GG12] and [JW16]. A huge set of examples of Belyi maps are available at the LMFDB which we explain in the final subsection.

7.1 A Genus 0 Example

Example 15. *As we mentioned earlier, trees are among the simplest (but very rich!) cases to consider as dessins. Trees with 2, 3 and 4 vertices (which are unique) come as dessins by the Belyi map $f : \mathbb{P}^1(\mathbb{C}) \to \mathbb{P}^1(\mathbb{C})$, $f(z) = z^n$ for $n = 2, 3, 4$. This map for any n gives a "star shape" dessin which is called "n-star" graph. As the equation is defined over $\mathbb{Q}$, we get a length 1 Galois orbit and the Monodromy group is generated by id and $(1, ..., n)$.*
The smallest non trivial cases in trees to consider, are two other trees with 4 edges: Considering the left hand side dessin, a simple calculation shows that the degree 4 polynomial corresponding to

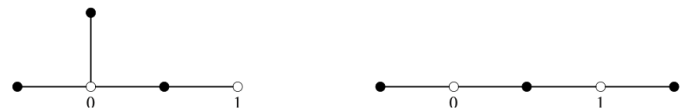


Figure 4: Non trivial trees with 4 edges

$\mathcal{D}$ is:

$$f_{\mathcal{D}}(z) = -\frac{256}{27} z^3 (z-1)$$

which is the Belyi function associated to this dessin. The permutation representation pair is given by $\sigma_0 = (1,2,3), \sigma_1 = (3,4)$.
Note that both dessins in figure 4 are particular cases of the configuration depicted in figure 5 , which we shall call the (m,n)-double star dessin, and whose corresponding Belyi function is of the form $f(z) = C_{m,n} z^m (z-1)^n$, where the constant $C_{m,n}$ can be determined by similar calculations.

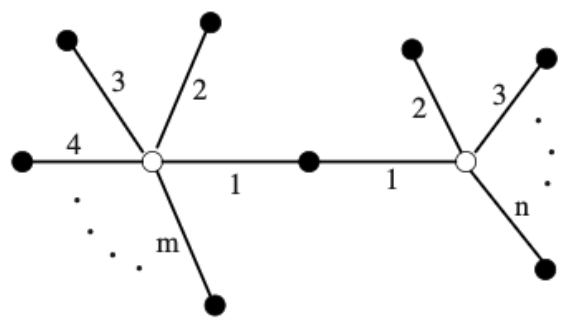


Figure 5: Double star dessins

The following proposition is not hard to prove ([GG12]):

Proposition 14. *The action of $\mathbb{G}$ on trees with less than five edges is trivial.*

7.2 A Genus 1 Example

Example 16. *Consider the elliptic curve X given by the equation*

$$X : y^2 = x(x-1)(x-\alpha),$$

where we take

$$\alpha := \frac{1}{\sqrt[3]{2}} \in \mathbb{R}$$

Clearly, this curve is defined over $\overline{\mathbb{Q}}$. We need a meromorphic function on X to go through Belyi's theorem, so we start with the projection

$$X \to \hat{\mathbb{C}}, (x, y) \mapsto x$$

which is ramified at the critical points $(\infty, \infty), (0, 0), (1, 0)$ and $(\alpha, 0)$, with critical values $\infty, 0, 1$ and α. In order to find a Belyĭ function, we need to eliminate the critical value α, so we compose this projection with the function $x \mapsto x^3$ which sends the algebraic number α to the rational number $\frac{1}{2}$, while fixing the other three critical values. To deal with the resulting critical value $\frac{1}{2}$, we now apply the function $\beta_{1,1} : z \mapsto 4z(1-z)$ which sends $\frac{1}{2}$ to 1, and sends 0,1 and ∞ to 0,0 and ∞. The composition

$$\beta : (x, y) \mapsto x \mapsto x^3 \mapsto 4x^3 \left(1 - x^3\right).$$

of these three steps therefore maps the critical values of the first step into $\{0, 1, \infty\}$ in the following way:

$$\infty \mapsto \infty, \quad 0 \mapsto 0, \quad 1 \mapsto 0, \quad \alpha \mapsto 1.$$

Using polynomials like this to deal with algebraic critical values may induce new ramifications. Here $x \mapsto x^3$ is ramified at $x = 0$ and $x = \infty$, so the composition $(x, y) \mapsto x^3$ is ramified over $0, 1, \infty$ and $\frac{1}{2}$. However, the last step is ramified only at ∞ and $\frac{1}{2}$, and sends these points to ∞ and 1 , so the composition β is a Belyĭ function $X \to \hat{\mathbb{C}}$. Considering inverse images of ∞, 0 and 1 with their multiplicities, we get the following dessin: Now, let's consider a non-trivia Galois

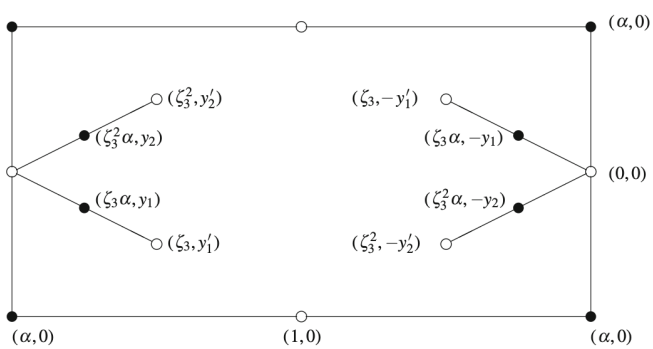


Figure 6: Dessin for Example 2

action $\sigma \in \mathbb{G}$:

$$\sigma(\sqrt[3]{2}) = \zeta_3^{-1}\sqrt[3]{2}$$

Then X^σ is given by

$$y^2 = x(x-1)\left(x - \frac{\zeta_3}{\sqrt[3]{2}}\right)$$

the Belyĭ function is $\beta(x, y) = 4x^3(1 - x^3)$, but the dessin now has the form given in next figure: Note that one has to identify opposite sides of the rectangle to get a dessin on a torus. The Belyĭ

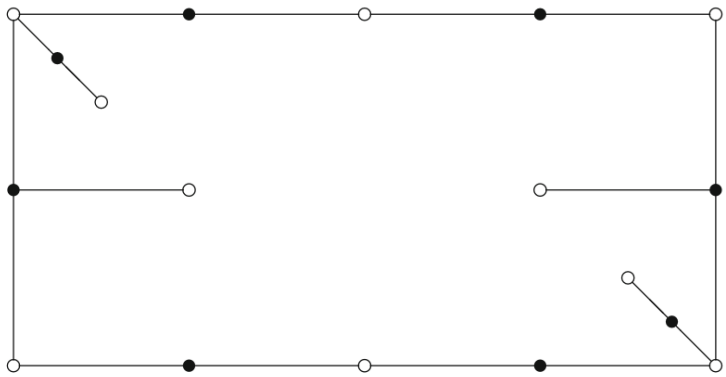


Figure 7: Galois conjugate of figure 6

function is given by the same formula as for the original dessin because it is a polynomial in $\mathbb{Q}[x]$.

7.3 Two Examples From LMFDB

The huge database of LMFDB[1] has been recently used to uncover and understand new patterns in number theoretic objects([He+25] and [SS25]). LMFDB also contains 1,111 examples of Belyi maps and highlights some interesting facts and examples which we breifly mention here.

Example 17. *LMFDB highlights this Belyi map as the lowest degree map (degree 4) with hyperbolic geometry type. The given Belyi map is*

$$f(x, t) = \frac{-4}{27}t$$

which is defined on the genus 1 curve (elliptic curve) $t^2 + (2x^2 - 4x + 9)t + x^4 = 0$. The curve is isomorphic with the minimal Weierstrass equation $y^2 = x^3 + x^2 + 16x + 180$ and there is also a page for this curve on LMFDB. This belyi map is defined on $\mathbb{Q}$ and the orbit size for the action of $\mathbb{G}$ is 1. The monodromy group for this map is S_4.

Example 18. *The longest Galois orbit on LMFDB belongs to a degree 7 Balyi map (with a huge formula!) which is defined on the curve (with a huge polynomial!) with genus 1. The Galois orbit contains 21 elements (and so the field of definition is a extension of degree 21 !)*

[1] L-functions and Modular Forms DataBase

8 Two Applications in Number Theory

In this final section, we mention to applications of Belyi maps and dessins in number theory. The first one is from [LZ03] and the second one is from [JW16].

8.1 ABC Conjecture

There exist several versions of the ABC problem; here we choose one of them which can be more relavant. Let a, b, c be three positive coprime integers such that $a + b = c$. Note that $c = \max(a, b, c)$. Let

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}, \quad b = q_1^{\beta_1} q_2^{\beta_2} \dots q_m^{\beta_m}, \quad c = r_1^{\gamma_1} r_2^{\gamma_2} \dots r_n^{\gamma_n}$$

be their factorizations into prime factors. Denote by R the product of all the prime factors of a, b, c taken only once, that is,

$$R = p_1 p_2 \dots p_k q_1 q_2 \dots q_m r_1 r_2 \dots r_n.$$

The conjecture affirms that R cannot be too small compared to c. More precisely: let $c = R^\alpha$, or, equivalently, $\alpha = \log c / \log R$.

Conjecture 1. *(abc). For any $\alpha_0 > 1$ there exist at most a finite number of triples (a, b, c) such that the corresponding value of the parameter α satisfies the inequality $\alpha > \alpha_0$.*

Note that if we replace $\alpha_0 > 1$ with 1 , the statement turns wrong:

Example 19. *Take the equality $1 + (2^{6n} - 1) = 2^{6n}$. The contribution of $a = 1$ and of $c = 2^{6n}$ to R is only 2, while $b = 2^{6n} - 1$ is divisible by $2^6 - 1 = 63$ and hence divisible by 9. Therefore the contribution of b is not bigger than $b/3$, and we conclude that $R < c$, so $\alpha > 1$.*

The next example shows what kind of consequences may be obtained from the conjecture.

Example 20. *Consider the equality $x^n + y^n = z^n$. Obviously,*

$$R \leq xyz < z^3;$$

therefore

$$\alpha = \frac{\log c}{\log R} > \frac{\log z^n}{\log z^3} = \frac{n}{3}.$$

Thus, the abc conjecture implies that all Fermat equations with $n \geq 4$ taken together may have at most a finite number of coprime solutions.

Keeping in mind the conjecture, it is interesting to look for triples a, b, c with the parameter α as big as possible. The world record up to now belongs to Eric Reyssat: $2 + 3^{10} \cdot 109 = 23^5$, with $\alpha = 1.62998$ (rounded).This triple was found by a brute force search. A number of papers were dedicated to a more systematic search: see, for example [Nit93b], [Nit93a], [BB94] (which lists a complete list of known instances with $\alpha > 1.4$), [Nit93b] (which lists a complete list of instances with the prime factors not greater than 13), etc. A rather long list of instances with $\alpha > 1.2$ was collected **by Noam Elkies** several years ago.
It is clear that in a "good" example the numbers a, b, c must be well factorizable in prime factors. Several authors proposed to use, in a search for good triples, Belyi functions defined over $\mathbb{Q}$. The polynomials constituting a Belyi function are already well factorized. Substituting rational numbers into them, we may, with a little bit of luck, find interesting examples. Let us illustrate it by taking one particular (and very simple) Belyi function:

Example 21. *We take the dessin of example 2.3.1 in [LZ03] though for technical reasons we have changed its position: we put the vertex of degree 1 at infinity, the vertex of degree 3, at 0, and the center of the face of degree 1, at 1. The corresponding Belyi function is*

$$f(x) = \frac{64x^3}{(x+9)^3(x+1)}, \qquad \textit{and} \qquad f(x) - 1 = -\frac{(x^2 - 18x - 27)^2}{(x+9)^3(x+1)}.$$

The above two equalities, with $x = a/b$, may be rewritten as

$$64a^3b + \left(a^2 - 18ab - 27b^2\right)^2 = (a+9b)^3(a+b)$$

Substituting $(a, b) = (-32, 23)$ we get, after rearranging the terms,

$$11^2 + 3^2 \cdot 5^6 \cdot 7^3 = 2^{21} \cdot 23$$

which gives the value of α equal to 1.62599. This is not the record, but this is the second best value known up to now. However, this triple was found before the search undertaken by N. Magot [Mag94].

In general, the above Belyi function has produced 43 triples with $\alpha > 1.2$; among them 11 triples were "new" (with respect to the list of Elkies as it was in 1996). The best among the "new" triples was

$$2^{15} \cdot 5 \cdot 67 + 11^2 \cdot 23^8 = 7^3 \cdot 3023^3, \quad \alpha = 1.35670\ldots$$

Several hundred "new" triples with $\alpha > 1.2$ and not present in the Elkies list were found by L. Habsieger.

8.2 Elliptic Curves and Complex Multiplication

We mentioned Elliptic curves in the beginning of this note (proposition 4) and end of section 2.1 We introduced Elliptic curves as tori $T := \mathbb{C}/\Lambda$. Multiplying by a non-zero complex number, we may assume that Λ is the lattice $\mathbb{Z} + \mathbb{Z}\tau$ with period quotient $\tau \in \mathbb{H}$. For any positive integer n, the holomorphic function $\mathbb{C} \to \mathbb{C}, z \mapsto nz$ induces an unramified covering $\alpha_n : T \to T$ which has degree $|\Lambda : n\Lambda| = n^2$. If there is a Belyi function $\beta : T \to \mathbb{P}^1(\mathbb{C})$ of degree d, then it's easy to show that $\beta \circ \alpha_n$ is also a Belyi function, now of degree $n^2 d$. We denote their respective dessins by $\mathcal{D}$ and $n^{-1}\mathcal{D}$, and call $n^{-1}\mathcal{D}$ an **unramified self-cover of $\mathcal{D}$ of degree** n^2. The following easy proposition shows that it is in fact reasonable to speak here about **coverings of dessins**:

Proposition 15. *Show that the dessin $n^{-1}\mathcal{D}$ has a group A of automorphisms isomorphic to $C_n \times C_n$, acting freely on vertices and faces, such that*

$$\mathcal{D} \cong A \backslash n^{-1}\mathcal{D}.$$

Similar to the coverings in topological context, it can be shown that self-covers are always unramified, and that the underlying base surface is isomorphic to the covering surface. One should note that such coverings exist only in genus 1:

Proposition 16. *For genus $g \neq 1$ there are no Riemann surfaces X with unramified coverings $X \to X$ of degree greater than 1.*

The essential point of our consideration is contained in the following:

Theorem 16. *For any dessin on a Belyi pair of genus 1 and for any positive integer n there is a self-cover of degree n^2.*

The algebraic background behind this theorem is the fact that elliptic curves always have $\mathbb{Z}$ as a ring of endomorphisms, where each $n \in \mathbb{Z}$ acts as the endomorphism induced by $z \mapsto nz$; here we again write the elliptic curve as a torus $T = \mathbb{C}/\Lambda$ and consider $n = 0$ as the trivial endomorphism sending the curve to the point 0 mod Λ. Note that the word endomorphism has to be understood here as indicating not only an endomorphism of the Riemann surface, but also an endomorphism of the group, fixing the point 0 mod Λ. Thus the covering transformation $z \bmod \Lambda \mapsto nz \bmod \Lambda$ inducing the quotient map $n^{-1}\mathcal{D} \to \mathcal{D}$ counts as an endomorphism, whereas the (Riemann surface automorphism) $z \bmod \Lambda \mapsto (z+a) \bmod \Lambda$ does not if $a \notin \Lambda$! This also helps to explain our slightly inconsistent notation, writing $\Gamma\backslash\mathbb{H}$ for quotients of the space $\mathbb{H}$, but $\mathbb{C}/\Lambda$ for quotients of the group $\mathbb{C}$, as is conventional in group theory.

If the endomorphism ring End T of T is strictly larger than $\mathbb{Z}$, we call T an **elliptic curve with complex multiplication**. This is the case if and only if the period quotient τ generates an imaginary quadratic extension K of $\mathbb{Q}$. If we denote the ring of integers of K by $\mathcal{O}_K$, and lift

the endomorphisms to $\mathbb{C}$, then it is easy to see that End T is the subring of all $\alpha \in \mathcal{O}_K$ such that $\alpha\Lambda \subseteq \Lambda$. An automorphism of T occurs if α is a root of unity; since roots of unity generate imaginary quadratic fields if and only if they are of order 3, 4 or 6, we have the following:

Theorem 17. *A dessin $\mathcal{D}$ of genus 1 belongs to an elliptic curve with complex multiplication by $\mathbb{Z}\left[e^{2\pi i/3}\right]$ (or $\mathbb{Z}[i]$) if $\mathcal{D}$ has an automorphism of order 3 (or 4, respectively) fixing at least one vertex or face.*

Section 10.2.3 of [JW16] discusses dessins with this condition.
It's even possible to give a complete condition from dessins on an elliptic curve to determine if it is CM or not (theorem 10.5 of [JW16]). Note that the following theorem may not be always computationally efficient!:

Theorem 18. *Let T be an elliptic curve defined over a number field (coefficients of T comes from an algebraic finite degree extension of $\mathbb{Q}$). T has complex multiplication if and only if every dessin on T has* ***infinitely many self-covers of prime degree****.*

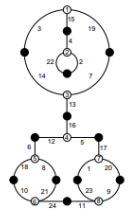
9 Some Research Directions

In this final section I will explain briefly two different research approaches regarding Belyi maps and dessin d'enfants (and their generalizations.)

9.1 Computation on Belyi maps and Dessins

As I mentioned in this note, computations on Belyi maps and dessins are not easy. One popular trend in this subject, as we mentioned in section 7 (on LMFDB examples) is to use machine learning (and deep learning) methods to find patterns in the objects we want to study.
one interesting example of this kind is investigated in [HHP21]. In this paper, authors gather a database of 191 dessins as concrete play-ground and they setup an examination: *can one "look" at a dessin and classify its degree of extension over* $\mathbb{Q}$*?* Indeed, a central theme of Grothendieck's Esquisse is to understand how dessins furnish a representation of the absolute Galois group which governs all algebraic extensions over rationals. Dessins in this paper are coming from **Extremal K3 surfaces** and **Seiberg-Witten Theory** in physics. Explicit calculations show all of them have extension degree less than 5.
Using a supervised method on the nurons of a neural network (from the **TensorFlow Keras Library**), output of each of the 4 neurons represents the probability that the dessin had a particular extension, taking the most probable extension as the predicted classification. Rather surprisingly, they find that a simple multi-layer perceptron with standard Leaky ReLU activation function and dense layers can classify this problem within the dataset, enhanced by permutation equivalence to a size of the order 10^5 to 92% accuracy. This is quite contrary to the fact how difficult it is to compute explicit Belyi maps and to the intuition built over the last couple of years how number-theoretical problems respond poorly to machine-learning. Perhaps, dessins are closer to complex analysis than to number theory.
An example of a dessin which is studied in this paper is the figure 8. Their data engineering method was to use "adjacent matrices" to represent a dessin as the input data for their model. In the conculsion of their research, they mention further questions to be investigated:
"There is an infinitude of dessins of varying complexity and it is known that $Gal(\bar{\mathbb{Q}}/\mathbb{Q})$ *acts even faithfully on genus 0 and genus 1 families thereof individually. Can our neural network extrapolate to these? Indeed, as a part of more ambitious plan, can we establish machine-learning models which help compute actual Belyi maps? Indeed, what about other aspects of dessins, such as cartographic groups and permutation triples, many of which have also found their place in theoretical physics; can these be machine-learnt?"*

$$\begin{pmatrix} 0 & 1 & 2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 \end{pmatrix}$$



{{{3,19,15},{4,2,22},
{14,7,13},{16,5,12},{17,20,1},
{6,8,18},{21,24,10},{23,9,11}},
{{24,11},{10,18},{8,21}, {1,23},{20,9},{17,5},
{12,6},{16,13},{14,3}, {19,7},{15,4},{22,2}}}

9-5-5-2-2-1 ($\mathbb{Q}$)

Figure 8: A dessin from [HHP21]

9.2 Grothendieck-Teichmuller Theory

Central theme of Grothendieck-Teichmüller Theory is to study $G_{\mathbb{Q}} = \mathbb{G} = Gal(\bar{\mathbb{Q}}/\mathbb{Q})$ via its actions on fundamental groups of geometric objects (varieties, schemes, stacks...). And above all, according to Grothendieck, one should concentrate attention on **the moduli spaces of curves**, which in some sense contain all possible information about algebraic curves and varieties over $\overline{\mathbb{Q}}$. Why this point of view is important? Actually, geometric Galois actions always stem from a basic situation:
if X is any algebraic variety (scheme, stack...) defined over $\mathbb{Q}$, let $\pi_1(X)$ denote its topological fundamental group and $\widehat{\pi}_1(X)$ its algebraic fundamental group, which is the profinite completion of the topological one. Then there is a canonical outer action

$$G_{\mathbb{Q}} \to \mathrm{Out}\,(\widehat{\pi}_1(X))$$

Moreover this outer action preserves conjugacy classes of some important objects (inertia groups). Let's explain where the outer Galois action comes from: The left-hand column of the following diagram shows a finite cover Y of X, sitting under the universal cover $\widetilde{X}$ of X, with Galois group the topological π_1, the middle column shows the function field situation over $\mathbb{C}$, where the top field is the compositum of all the function fields of the finite covers Y and therefore the Galois group is the profinite completion of the topological π_1, and the right-hand column uses the Lefschetz theorem to descend from $\mathbb{C}$ to the algebraically closed subfield $\overline{\mathbb{Q}}$ without changing the Galois group, so that the natural inclusion of the field $\mathbb{Q}(X)$ into $\overline{\mathbb{Q}}(X)$, with Galois group $\mathbb{G}$, gives a canonical outer action of $\mathbb{G}$ on $\widehat{\pi}_1(X)$. So, following the program Grothendieck sketched out in his famous text Esquisse d'un Programme, one should begin by examining the action of $\mathbb{G}$ on fundamental groups of moduli spaces of curves, obtaining in this way information about the elements of $\mathbb{G}$ which is very different from what can be obtained by studying number fields directly.

$$\begin{array}{ccc} \widetilde{X} & \widetilde{\mathbb{C}(X)} & \widetilde{\overline{\mathbb{Q}}(X)} \\ | & | & | \\ \pi_1(X)\Big(\; Y & \widehat{\pi}_1(X)\Big(\; \mathbb{C}(Y) & \widehat{\pi}_1(X)\Big(\; \overline{\mathbb{Q}}(Y) \\ | & | & | \\ X & \mathbb{C}(X) & \overline{\mathbb{Q}}(X) \\ & & G_{\mathbb{Q}} \; (\; | \\ & & \mathbb{Q}(X) \end{array}$$

Figure 9: Outer Galois Action

The very first question one might ask about an element $\sigma \in \mathbb{G}$ is: How can you write it down? How can you express it, say what it is? The fact is that $\mathbb{G}$ is a profinite group. Recall that the profinite completion of a group is given by the inverse limit of the system of all its finite quotients: $\widehat{G} = \lim_{\leftarrow} G/N$ where N runs through the normal subgroups of finite index of G. A profinite group is the inverse limit of its finite quotients, so that to give an element of $\mathbb{G}$ explicitly, we need to know its image in each finite quotient. But we do not even know the finite quotients of $\mathbb{G}$. The only element which can really be written down in an explicit sense is complex conjugation.
The purpose of the theory developed by Grothendieck in the second section of Esquisse d'un Programme is:

1. to identify each element $\sigma \in G_{\mathbb{Q}}$ with a pair

$$(\chi(\sigma), f_\sigma) \in \widehat{\mathbb{Z}}^* \times \widehat{F}_2'$$

   Here $\chi : \mathbb{G} \to \widehat{\mathbb{Z}}^*$ is just the cyclotomic character giving the action of $\mathbb{G}$ on roots of unity; we have the exact sequence

$$1 \to G_{\mathbb{Q}^{\mathrm{ab}}} \to \mathbb{G} \to \widehat{\mathbb{Z}}^* \to 1$$

   so for any $\sigma \in \mathbb{G}, \chi(\sigma)$ is a very well-understood quantity. As for $f_\sigma \in \widehat{F}_2'$, which is the derived subgroup of the profinite completion of the free group $\widehat{F}_2$ on two generators. $\widehat{F}_2$ is a much simpler group than $\mathbb{G}$. It is also profinite, so we still cannot always write down an element completely. However, because we understand this group well, there are various ways to characterize an element or a certain subset of elements of $\widehat{F}_2$ just by giving its properties. This is why it is very interesting to be able to write $\sigma \in \mathbb{G}$ as a pair $(\chi(\sigma), f_\sigma)$ both of whose components belong to much simpler groups than $\mathbb{G}$.
   Grothendieck indicated how to do this in Esquisse d'un Programme, but the work was

completed by Drinfeld and Ihara.

2. The second part of Grothendieck's program is still the hard part: Find necessary and sufficient conditions on $f \in \widehat{F}_2'$ for it to come from a $\sigma \in \mathbb{G}$.
   Studying the action of $\mathbb{G}$ on fundamental groups of moduli spaces of curves has yielded several beautiful necessary conditions. But sufficiency is still a real mystery.

There are many interesting open problems in this theory and [LS09] is a reference for these problems.

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